Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting*

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Abstract

The direction of the asymmetric volatility response—measured by the GJR-GARCH γ parameter—varies systematically across asset classes in ways that have direct implications for model selection, risk management, and portfolio construction. Using daily data for seven primary assets (equities, gold, bonds, emerging markets, cryptocurrency) over 2017–2026, extended to 26 assets in validation analyses, we make two contributions. First, we propose a leverage direction taxonomy: equities exhibit standard leverage ($\gamma > 0$), gold displays regime-dependent leverage (inverted during fear-driven rallies, standard during liquidation-driven declines), and bonds show near-zero asymmetry. This taxonomy determines optimal GARCH specification: GJR-GARCH significantly outperforms symmetric GARCH only when $|\gamma| > 0.10$, correctly classifying all nine Diebold-Mariano comparisons in the primary sample, with 6/6 correct out-of-sample predictions. Second, we link leverage direction to the economic channel of volatility targeting (VT) alpha: γ predicts whether VT acts as trend-following or contrarian within homogeneous equity markets (Spearman $\rho = 0.886$, $p = 0.019$ for six equity-type assets), but this relationship does not extend to diverse asset classes ($\rho = -0.448$, $p = 0.14$ for twelve assets). Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across

*I thank seminar participants for helpful comments. All errors remain my own. Data and replication code available upon request.

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all tested assets and driven by base volatility level ($\rho = 0.944$ for primary assets, $\rho = 0.83$ for the extended $N = 14$ sample), independent of γ. These findings delineate a complexity ceiling: increasing model sophistication (DCC-GARCH, copula, GARCH-MIDAS, Markov-switching) improves statistical measurement but not investable allocation decisions.

Keywords: GARCH, leverage effect, gold, asymmetric volatility, Value-at-Risk, Basel III, volatility targeting, cross-asset, regime-dependent

JEL Classification: C22, C53, G01, G11, G17

1 Introduction

The GARCH family of models (Bollerslev, 1986) remains the workhorse for daily volatility forecasting in financial markets. A substantial literature has developed around asymmetric extensions—most notably the GJR-GARCH of Glosten, Jagannathan, and Runkle (1993) and the EGARCH of Nelson (1991)—designed to capture the empirical observation that negative equity returns tend to increase future volatility more than positive returns of equal magnitude. This ”leverage effect,” first noted by Black (1976) and formalized by Christie (1982), is well-documented for equity markets and has become a standard consideration in volatility model specification.

However, the implicit assumption in much of the applied GARCH literature—that asymmetric models uniformly improve upon their symmetric counterparts—warrants closer examination when moving beyond equities. While Chevallier and Ielpo (2017) document inverted asymmetric volatility in gold and several agricultural commodities, the implications of this reversal for model selection, risk management, and portfolio construction have not been systematically explored.

This paper makes two contributions to the volatility forecasting and financial risk management literature.

First, we propose a leverage direction taxonomy based on the GJR-GARCH γ parameter across seven primary assets spanning equities (SPY, QQQ, EEM), precious metals (GLD, SLV), government bonds (TLT), and cryptocurrency (BTC-USD), validated on up to 26 assets. The sign of γ corresponds to the asset's price-driving mechanism: risk assets exhibit standard leverage ($\gamma > 0$), safe-haven assets show inverted leverage ($\gamma < 0$), and interest rate instruments display no significant asymmetry ($\gamma \approx 0$). Gold's mean γ is statistically indistinguishable from zero in the canonical 2010–2026 replication (mean +0.002, HAC $t = +0.15$, 67% of quarterly windows negative), but it is sharply regime-dependent: bull periods exhibit $\gamma < 0$ (inverted leverage, consistent with fear-driven safe-haven flows) while bear periods exhibit $\gamma > 0$ (standard leverage, consistent with liquidation-driven declines), with the regime difference statistically significant ($t = -3.79$, $p < 0.001$). The two regimes offset over the full sample, producing a near-zero average that masks an economically meaningful regime structure. This taxonomy determines optimal GARCH specification: GJR-GARCH significantly outperforms symmetric GARCH only when $|\gamma| > 0.10$. The $|\gamma| > 0.10$ rule correctly classifies all nine Diebold-Mariano comparisons in the *primary* sample (in-sample, Table 3) and 6/6 comparisons out-of-sample, using 2023 γ estimates to predict 2024–2025 model choice. We propose that γ direction, rather than the commonly used return skewness, provides a more reliable model selection criterion.

Second, we apply the leverage taxonomy to volatility targeting (VT), linking leverage direction to the economic channel through which VT generates alpha. This generalizes the equity-specific finding of Hood and Raughtigan (2025). Within equity-type assets, γ predicts whether VT acts as trend-following or contrarian (Spearman $\rho = 0.886$, $p = 0.019$ for six equity-type assets; see Section 5.5 for domain restrictions), with out-of-sample predictive power ($\rho = 0.821$ using γ from 2010–2017 to predict trend beta in 2018–2026). However, this relationship does not extend to diverse asset classes ($\rho = -0.448$, $p = 0.14$ for twelve assets), delineating the proposition's domain. Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across all twelve tested assets and driven by base volatility level (Pearson $\rho = 0.944$, $p < 0.001$, $N = 5$ primary assets; Pearson $\rho = 0.83$, $p = 0.0002$, $N = 14$ in the extended sample), independent of γ.

Supporting findings reinforce a broader pattern: increasing model sophistication improves statistical measurement but not investable allocation decisions. Supplementary evidence of cross-market information transmission is presented in Appendix B.

The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 describes data and methodology. Section 4 presents empirical results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Literature Review

2.1 GARCH Models and the Leverage Effect

Bollerslev's (1986) GARCH model captures the well-documented volatility clustering in financial returns but assumes symmetric responses to positive and negative innovations. The "leverage effect"—negative equity returns increasing future volatility more than positive returns of equal magnitude—was first noted by Black (1976) and attributed to changes in financial leverage by Christie (1982). This motivated asymmetric extensions: Nelson (1991) introduced the EGARCH, modeling the logarithm of conditional variance with separate coefficients for sign and magnitude, while Glosten, Jagannathan, and Runkle (1993) proposed the GJR-GARCH, adding an indicator function for negative innovations. Engle and Siriwardane (2018) provide a structural interpretation linking GJR's γ to capital structure dynamics. Two parallel extensions widened the GARCH design space. Engle et al. (2013) introduce GARCH-MIDAS, which decomposes conditional variance into a high-frequency GARCH component and a low-frequency macro-driven component, tying long-run volatility to observable economic fundamentals. Patton and Sheppard (2015) show that the asymmetric response of volatility can be traced specifically to realized negative semivariance ("bad" volatility), sharpening the interpretation of γ-type asymmetry as a downside-risk channel. Asymmetric models generally improve upon symmetric specifications for equity indices (see Hansen and Lunde 2005 for a comprehensive comparison of 330 GARCH variants), but whether the direction of the asymmetric response extends systematically to non-equity assets—and whether the "bad"-volatility interpretation reverses for safe havens—has received less attention.

2.2 Commodity Volatility and Inverted Asymmetry

The behavior of volatility in commodity markets differs from equities in important ways. Chevallier and Ielpo (2017) investigate the leverage effect across a broad set of commodities and find that gold, wheat, coffee, and cocoa exhibit inverted asymmetric volatility—where positive price shocks increase volatility more than negative shocks—consistent with earlier findings for precious metals (Batten et al., 2010).

The economic interpretation differs from equities: while equity leverage operates through capital structure, commodity price increases during stress reflect uncertainty about supply, demand, or financial conditions. For gold specifically, Baur and McDermott (2010) and Baur and Lucey (2010) establish that gold functions as both a hedge and a safe haven for developed market equities, with hedging effectiveness increasing during extreme downturns. Chang et al. (2021) use a Markov-switching GJR-GARCH to link gold's inverted asymmetry to high-volatility regimes, showing that gold's safe-haven ability is concentrated in these regimes. Our analysis extends this to a broader asset class framework, examining implications for model selection, VaR compliance, and portfolio construction—dimensions not addressed in the existing literature.

2.3 Value-at-Risk and Basel III

The Basel Committee's internal models approach (BCBS, 2006, 2019) requires banks to backtest their VaR models against realized losses. Kupiec (1995) provides the unconditional coverage test, while Christoffersen (1998) adds the independence dimension. McNeil, Frey, and Embrechts (2015) provide a comprehensive treatment of quantitative risk management methodology.

The distributional assumption significantly impacts VaR accuracy. Bollerslev (1987) introduced Student-t innovations for GARCH models, and subsequent work (e.g., Hansen 1994) developed skewed-t and other flexible distributions. Kuester, Mittnik, and Paolella (2006) find that GARCH with appropriate distributions outperforms historical simulation and extreme value approaches.

2.4 Volatility Targeting

Moreira and Muir (2017) demonstrate that scaling portfolio exposure by the inverse of conditional variance—volatility targeting—generates significant alphas across equity factors, value, momentum, and currency carry. Their key insight is that changes in volatility are not offset by proportional changes in expected returns, creating a risk-return tradeoff that VT exploits.

Harvey, Hoyle, Korgaonkar, Rattray, Sargaison, and Van Hemert (2018) extend these results to multi-asset portfolios. Fleming, Kirby, and Ostdiek (2001, 2003) provide earlier evidence that volatility timing improves performance. Hood and Raughtigan (2025) demonstrate that equity VT alpha arises primarily from implicit trend-following via the leverage effect: because negative returns raise volatility and trigger deleveraging, VT mechanically reduces exposure after declines. This equity-specific mechanism reverses for inverted-leverage assets, connecting directly to our γ taxonomy.

Cederburg et al. (2020) show that using VIX as the scaling signal produces approximately 4.9% alpha versus realized-variance scaling. Bozovic (2024) confirms VIX dominance through forward-looking information. DeMiguel, Martin-Utrera, and Uppal (2024) reframe volatility-managed portfolios from a multifactor perspective, showing that the in-sample Sharpe gains of VT strategies are substantially attenuated once exposures to standard risk factors are controlled for—motivating the leverage-direction cross-section we develop here. Xu (2024) validates real-time viability across 197 equity factors. However, VT effectiveness across different leverage regimes—particularly for inverted-leverage assets—has not been previously examined.

2.5 Deep Learning for Volatility Forecasting

Recent work has explored neural network approaches to volatility forecasting. Hybrid GARCH-LSTM models (e.g., Kim and Kim, 2019; Araya et al., 2024) attempt to combine the structural advantages of GARCH with the flexibility of deep learning. However, the incremental improvement over well-specified GARCH models remains debated, particularly at daily frequency where GARCH models may already extract the available signal in squared returns (see Bucci 2020 for a skeptical assessment).

3 Data and Methodology

3.1 Data

We use daily adjusted closing prices for seven exchange-traded assets: SPDR S&P 500 ETF Trust (SPY), Invesco QQQ Trust (QQQ), iShares MSCI Emerging Markets ETF (EEM), SPDR Gold Shares (GLD), iShares Silver Trust (SLV), iShares 20+ Year Treasury Bond ETF (TLT), and Bitcoin (BTC-USD). Data are sourced from Yahoo Finance adjusted close prices for the period January 2017 through March 2026.

Data source justification. Yahoo Finance is among the most widely used free data sources in empirical finance research. For large-cap U.S. equities and highly liquid ETFs—the asset universe of this study—prior validation studies document high concordance between Yahoo Finance and institutional-grade databases. Bali et al. (2016) employ Yahoo Finance as the primary data source for their comprehensive textbook on empirical asset pricing methods, establishing an accepted precedent in the literature. Prior comparisons of CRSP with free alternative data sources—including Yahoo Finance—document that discrepancies are concentrated in small-cap stocks subject to infrequent trading and delayed corporate action adjustments, whereas large-cap U.S. equities and highly liquid ETFs track institutional-grade feeds closely. Neither concern applies to our ETF sample, where average daily dollar volume exceeds $1 billion for all instruments except SLV.

For U.S.-listed ETFs, Yahoo Finance prices reflect official exchange data with corporate action adjustments (splits, dividends). We perform three validation checks: (i) spot-checking price accuracy against Bloomberg terminal data for key crisis dates (COVID crash, March 2020; Iran/Hormuz crisis, March 2026), finding no discrepancies; (ii) verifying that VIX index values from Yahoo Finance match CBOE official data obtained via FRED (correlation > 0.9999 across 2,200+ trading days); and (iii) confirming that computed daily return distributions match those reported in published studies using CRSP data for overlapping periods (Moreira and Muir, 2017; Cederburg et al., 2020).

While Yahoo Finance data may contain occasional errors—for example, delayed corporate action adjustments for less liquid securities or temporary data feed interruptions—these issues are negligible for the large-cap, highly liquid ETFs in our sample. We nevertheless acknowledge this as a limitation: all results should be verified with institutional-grade data (CRSP, Bloomberg, or Refinitiv) before deployment in production risk management systems. Daily returns are computed as simple percentage changes $r_t = (P_t - P_{t-1})/P_{t-1}$.

Table 1 reports descriptive statistics. All return series reject normality (Jarque-Bera p < 0.001), exhibit significant ARCH effects (Engle's LM test p < 0.001), and are stationary (ADF p < 0.001). Excess kurtosis ranges from 3.52 (GLD) to 14.61 (SPY), motivating the consideration of fat-tailed distributions for VaR estimation.

Sample period justification. Our primary asset data span January 2017 through March 2026 (approximately 2,300 trading days per asset), partitioned into three non-overlapping windows for all in-sample-versus-out-of-sample analyses: in-sample training (2017-01 to 2022-12), main out-of-sample (2023-01 to 2024-12), and out-of-sample validation (2025-01 to 2026-03). We acknowledge that an eight-year primary sample is shorter than the multi-decade panels common in the GARCH forecasting literature. However, several design features mitigate this limitation.

First, the rolling estimation window ($w = 504$) is re-estimated at every forecast origin, so each model fit draws on the most recent two years of data—a standard requirement for GARCH parameter stability (Hwang and Valls Pereira, 2006). For extended-window robustness checks ($w \in \{2000, 3000, 5000\}$), training data reach back to 2006 or earlier, encompassing the Global Financial Crisis and subsequent recovery. The estimation sample therefore spans 11+ calendar years even when the evaluation window is shorter.

Second, our out-of-sample evaluation periods satisfy the minimum length recommendation of Hansen and Lunde (2005), who argue that at least 252 trading days are necessary for reliable forecast comparison. Our primary OOS (2023–2024) contains approximately 504 days, and the validation OOS (2025–March 2026) adds a further $\sim$300 days. Critically, these periods cover distinct market regimes: the post-COVID normalization and low-volatility environment of 2023, the rate-cut cycle and renewed equity momentum of

2024, and the geopolitical shock of the 2025–2026 Iran/Hormuz crisis—providing meaningful regime diversity for evaluating model robustness.

Third, the central research question concerns the *sign* of the leverage parameter γ—a structural characteristic of each asset's volatility dynamics—rather than short-horizon return prediction. Structural features such as the leverage effect are detectable in relatively short samples: Glosten et al. (1993) established the leverage effect in U.S. equities using fewer than 3,000 daily observations, and Engle (2004) documented asymmetric variance dynamics for individual stocks with as few as 500 observations per estimation window. The gamma taxonomy documented in this paper—positive for equities, negative for gold, near-zero for bonds—is invariant across all window sizes tested ($w \in \{252, 504, 756, 1000, 2000, 3000, 5000\}$; Table 10), confirming that this structural classification does not require longer samples.

Fourth, we verify that the key results—the gamma taxonomy, the complexity ceiling, and the VaR orthogonality finding—hold in the high-volatility sub-period of 2020–2022 (COVID crash through rate-hiking cycle), which represents the most stressed market conditions in our sample. Results are qualitatively unchanged (details available upon request), further supporting the robustness of our conclusions across market regimes.

3.2 Volatility Models

We compare the symmetric GARCH(1,1) of Bollerslev (1986), $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, against the GJR-GARCH(1,1) of Glosten et al. (1993):

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}_{\varepsilon_{t-1}<0})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{1}$$

where γ captures asymmetric responses: $\gamma > 0$ implies standard leverage (negative returns increase variance), $\gamma < 0$ implies inverted leverage.

Mean specification. We adopt the zero-mean specification $r_t = \sigma_t\varepsilon_t$ throughout. This choice is grounded in the well-documented finding that the conditional mean of daily equity returns is negligible relative to conditional variance: for a typical daily mean of

0.04% and daily standard deviation of 1%, the mean-to-volatility ratio is approximately 0.04, making its contribution to variance dynamics immaterial (Bollerslev, 1986; Hansen and Lunde, 2005). Francq and Zakoïan (2004) show that quasi-maximum likelihood estimation of GARCH parameters remains consistent regardless of mean specification when the true conditional mean is small relative to conditional variance. As a robustness check, we re-estimate all models with an AR(1) mean specification $r_t = \mu + \phi r_{t-1} + \sigma_t \varepsilon_t$; results are qualitatively unchanged—QLIKE rankings, gamma sign classifications, and DM test conclusions are all preserved (Section 4.6, Model Specification).

All models are estimated with Normal quasi-likelihood via the `arch` package (Sheppard, 2023). For VaR, we additionally consider Student-t innovations with fixed degrees of freedom.

3.3 Rolling Estimation

We employ a rolling window approach with re-estimation at each forecast origin. The primary window size is $w = 504$ trading days (approximately 2 calendar years), which satisfies the minimum sample size recommendation for GARCH estimation (≥ 500 observations; see Hwang & Valls Pereira, 2006) while avoiding contamination from distant regime changes. Robustness checks with $w \in \{252, 1000, 2000, 3000, 5000\}$ (Table 10) confirm that QLIKE rankings are qualitatively robust to window choice, though $w = 504$ produces the best QLIKE during high-volatility periods (2020–2021) while larger windows ($w \geq 2000$) perform marginally better during calm periods (2023–2024). The gamma sign classification is invariant to window size across all tested values.

For each out-of-sample date t, we estimate the model using data from $t - w$ to $t - 1$, and produce a one-step-ahead variance forecast $\hat{\sigma}_t^2$. The realized variance proxy is the squared daily return r_t^2.

3.4 Evaluation Criteria

3.4.1 Statistical Loss Functions

The primary evaluation metric is the QLIKE loss function (Patton, 2011), defined as the Gaussian quasi-log-likelihood contribution:[1]

$$\text{QLIKE} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t^2}{\hat{\sigma}_t^2} + \ln \hat{\sigma}_t^2\right) \tag{2}$$

QLIKE is robust to noise in the realized variance proxy and ranks forecasts consistently regardless of the proxy used, provided the proxy has equal bias across models.

3.4.2 Diebold-Mariano Test

We test the null hypothesis of equal predictive accuracy between models using the Diebold-Mariano test (Diebold & Mariano, 1995) with Newey-West HAC standard errors (5 lags):

$$DM = \frac{\bar{d}}{SE_{NW}(\bar{d})} \sim N(0,1) \tag{3}$$

where $d_t = L(r_t^2, \hat{\sigma}_{1,t}^2) - L(r_t^2, \hat{\sigma}_{2,t}^2)$ is the loss differential using QLIKE.

3.4.3 VaR Backtesting

Value-at-Risk at confidence level α is computed as:

- **Normal:** $\text{VaR}_\alpha = \Phi^{-1}(\alpha) \cdot \hat{\sigma}_t$
- **Student-t:** $\text{VaR}_\alpha = t_\nu^{-1}(\alpha) \cdot \sqrt{(\nu-2)/\nu} \cdot \hat{\sigma}_t$

where ν is the degrees of freedom (fixed at 5 in our baseline, consistent with typical empirical estimates for financial returns). The choice of df = 5 is validated out-of-sample: Section 4.4 demonstrates that Basel III Green Zone compliance is achieved for all df $\leq$

[1] This formulation is equivalent to $-\ln f(r_t \mid \hat{\sigma}_t^2)$ under Gaussian quasi-likelihood, up to an additive constant. Values are negative for daily returns because $\ln \hat{\sigma}_t^2 \approx \ln(10^{-4}) \approx -9.2$. **Lower (more negative) values indicate better forecasts.** An alternative convention in the literature defines the Patton (2011) centered loss as $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which produces values centered near 1.0. Both formulations preserve forecast rankings; we adopt the quasi-log-likelihood form throughout Tables 5–10 for consistency with maximum likelihood estimation.

6, and that jointly estimating df produces *worse* performance by over-adapting to quiet markets. The fixed df = 5 is thus a conservative choice that is robust across the empirical range df ∈ [4, 7].

We apply Kupiec's (1995) unconditional coverage test (LR statistic, χ^2 with 1 d.f.) and Christoffersen's (1998) independence test to assess violation clustering. Annual violation counts are classified per the Basel III traffic light system (Green: 0–4, Yellow: 5–9, Red: ≥10 violations per approximately 250 trading days).

3.4.4 Leverage Direction Analysis

To analyze the temporal stability of the leverage direction, we estimate GJR-GARCH on semi-overlapping quarterly windows (504-day windows advanced by 63 trading days) and collect the gamma estimates.

- **H0:** $E[\gamma] \geq 0$ vs. **H1:** $E[\gamma] < 0$ (inverted leverage)

using a one-sided t-test on the quarterly gamma series.

3.5 Volatility Targeting

Following Moreira and Muir (2017), the volatility-managed portfolio weight is:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t} \tag{4}$$

We set $\sigma_{\text{target}} = 10\%$ annualized for the GARCH-based analysis, and $\sigma_{\text{target}} = 12\%$ for the VIX-based analysis (Section 5.9), reflecting the VIX's systematic upward bias from the embedded variance risk premium. Weights are smoothed with a 5-day moving average and clipped to $[0, 1.5]$. Post-hoc transaction cost analysis (Section 5.1) confirms that the estimated annual turnover of 756% generates cost drag of only 8–15 basis points at typical SPY bid-ask spreads.

4 Empirical Results

4.1 Data Characteristics

Table 1 summarizes the distributional properties of daily returns for our seven assets over the full sample period (January 2017 to December 2025); the sample is partitioned into an in-sample training window (2017-01 to 2022-12), a main OOS window (2023-01 to 2024-12), and an OOS validation window (2025-01 to 2026-03), as described in Section 3. All return series exhibit significant excess kurtosis, ranging from 3.52 (GLD) to 14.61 (SPY), and decisively reject normality (Jarque-Bera $p < 0.001$). The ARCH LM test confirms significant conditional heteroskedasticity in all series ($p < 0.001$), justifying GARCH-family models.

Return skewness varies across assets: SPY (-0.315), QQQ (-0.189), GLD (-0.296), and EEM (-0.584) show negative skewness, while TLT ($+0.155$) is approximately symmetric. A naive model selection approach based on skewness would suggest asymmetric GARCH for all negatively skewed assets. As we demonstrate in Section 4.2, this criterion is misleading for gold: despite negative skewness, gold's conditional variance responds asymmetrically to *positive* shocks.

Annualized volatility spans an order of magnitude, from 14.6% (GLD) to 57.3% (BTC-USD), motivating our analysis of VT effectiveness as a function of base volatility (Section 4.5). The extreme minimum returns—SPY at -10.94% (March 2020 COVID crash) and BTC-USD at -37.17%—underscore the importance of fat-tailed distributions for VaR estimation (Section 4.4).

4.2 Leverage Direction Across Asset Classes

4.2.1 Rolling Gamma Estimates

We estimate the asymmetry parameter γ of GJR-GARCH(1,1) using a rolling window of 504 trading days, updating quarterly. Table 2 reports cross-sectional statistics for each asset.

The most striking finding is the **consistent sign pattern across asset classes**. For equities (SPY, QQQ, EEM), γ is uniformly positive across all quarterly estimates, indicating the well-documented leverage effect. In contrast, gold (GLD)'s rolling γ estimates straddle zero: the canonical K903 504/63 replication over 2010-01 to 2026-04 ($n = 58$ quarterly windows) yields a mean $\gamma = +0.002$ (std $= 0.055$) with HAC $t = +0.15$, statistically indistinguishable from no-asymmetry, while 67% of individual quarterly estimates remain below zero.[2] Gold's mixed-sign behavior contrasts sharply with the uniformly positive equity γ and reflects regime-dependent leverage (Section 4.2 below); it is consistent with prior evidence that gold acts as a fear-rally safe haven (Baur and McDermott, 2010; Baur and Lucey, 2010) but switches to standard leverage during liquidation-driven bear markets.

Treasury bonds (TLT) show near-zero γ (mean $= -0.008$, std $= 0.048$), with approximately equal proportions of positive and negative estimates, indicating no significant asymmetry. Bitcoin (BTC-USD) displays mild standard leverage (mean $\gamma = +0.117$), though with higher instability (std $= 0.136$) and occasional sign reversals.

Gold's mixed-sign behavior is consistent with two coexisting price-formation channels. During market stress, fear-driven buying drives prices upward, creating heightened uncertainty that elevates conditional variance and produces $\gamma < 0$—the safe-haven channel (Baur and McDermott, 2010; Baur and Lucey, 2010). During liquidation-driven gold bear markets, falling prices increase downside risk, the same mechanism as equity leverage, producing $\gamma > 0$ (Black, 1976; Christie, 1982). Over the full 2010–2026 sample these regimes offset, yielding a mean γ near zero (Table 4). Section 4.2 documents the regime split formally and shows that the bull–bear γ difference for gold is statistically significant ($t = -3.79$, $p < 0.001$), even though the unconditional mean is not. The absence of asymmetry in bonds, by contrast, reflects a single economic channel: interest rate changes do not systematically alter sovereign credit risk in either direction.

[2] An earlier draft of this paper reported mean $\gamma = -0.067$ (std $= 0.044$, HAC $t = -5.79$) with 93% of windows negative; a forensic audit (`paper/leverage-direction/errata_gld_rolling_gamma_forensic.md`, 2026-05-30) traced this estimate to an obsolete spec whose source experiment could not be re-located in the current ledger, and was unable to reconcile it with the K903 canonical estimate using the paper's stated 504/63 window. Table 4 and the discussion below adopt the K903 canonical numbers; the qualitative GLD finding (mixed-sign, regime-dependent behavior) survives this revision, but the headline "inverted leverage is statistically significant" claim does not.

4.2.2 Implications for Model Selection

Table 3 presents GARCH(1,1) versus GJR-GARCH(1,1) QLIKE comparisons across all asset-period combinations. The results reveal a clear pattern: GJR-GARCH significantly outperforms only when γ is consistently positive and above approximately 0.10.

For SPY ($\gamma \approx +0.21$), GJR achieves significantly lower QLIKE (DM $p = 0.001$ for 2023–2024, $p = 0.029$ for 2025). For QQQ in 2025 ($\gamma \approx +0.17$), GJR likewise wins significantly ($p = 0.023$). However, for GLD ($\gamma \approx 0$ on average; Table 4), TLT ($\gamma \approx 0$), and BTC-USD ($\gamma \approx +0.08$), Diebold-Mariano tests fail to reject equal accuracy at the 5% level across all tested periods.

This finding has practical implications: the conventional approach of selecting asymmetric GARCH models based on return skewness can be misleading. GLD exhibits substantial negative skewness (-0.296) yet shows no consistent leverage asymmetry in the canonical sample ($\gamma \approx 0$ on average; Table 4)—skewness-based selection would incorrectly prescribe GJR despite its lack of forecasting benefit for gold (Table 5). We propose that **the sign and magnitude of estimated** γ should replace skewness as the primary criterion.

4.2.3 Robustness

The γ sign classification is robust to window size (results qualitatively identical with $w = 252$ and $w = 756$), estimation period (2007–2026 for SPY shows the same pattern), and model specification (EGARCH leverage parameter shows consistent sign alignment with GJR γ).

Because our quarterly gamma estimates use overlapping windows (504-day windows stepped by 63 days, yielding 87.5% overlap between consecutive estimates), serial dependence in the gamma series is a first-order concern. As Harri and Brorsen (2009) emphasize, overlapping observations inflate the effective sample size relative to the number of independent data points, biasing naive t-statistics upward. We address this in three ways.

First, we employ Newey-West HAC standard errors (Newey and West, 1987) with

8 lags (chosen as $\lceil 0.75 \times T^{1/3} \rceil$ where T is the number of quarterly gamma estimates), which accounts for the moving-average structure induced by the overlap. Under the K903 canonical 504/63 rolling spec the HAC-corrected t-statistic for gold's mean γ is $t = +0.15$ (not significant), confirming that the unconditional GLD asymmetry, when correctly estimated, is statistically indistinguishable from zero. The HAC corrections for SPY and EEM remain strongly positive ($t = +11.1$ and $+11.9$ respectively under the same spec), confirming that the equity leverage finding is independent of the GLD revision and is itself robust to the overlap-induced inflation of naïve standard errors.

Second, we verify with a block bootstrap (10,000 replications, block length = 4 quarters to match the overlap structure). Under the K903 canonical spec, the bootstrap 95% confidence interval for gold's mean γ comfortably includes zero, consistent with the HAC inference of $t = +0.15$. The same procedure produces bootstrap intervals strictly bounded away from zero for SPY and EEM, again localizing the equity leverage finding to those assets and away from gold.

Third, we repeat the cross-sectional analysis using strictly non-overlapping 5-year windows (2006–2010, 2011–2015, 2016–2020, 2021–2025). While the smaller number of cross-sections ($N = 4$) limits formal inference, the gamma rankings are qualitatively preserved for equities and bonds: SPY exhibits positive mean γ in all four windows (range: +0.14 to +0.27), and TLT remains near zero in all windows ($|\bar{\gamma}| < 0.02$). For gold, the non-overlapping check is consistent with the K903 canonical evidence of regime-dependent leverage: GLD's mean γ is negative in the safe-haven-dominated windows (notably 2008–2012 and 2021–2025) but turns positive in the gold-bear window (2013–2015, $\bar{\gamma} \approx +0.20$; see Section 4.2). The sign-based taxonomy—positive for equities, near-zero on average but regime-dependent for gold, near-zero for bonds—is thus robust to complete elimination of window overlap.

We conclude that while the 87.5% overlap in our primary analysis inflates the number of gamma observations, it does not drive the qualitative findings. The overlap provides finer temporal resolution for regime analysis (Section 5.5), while the non-overlapping check confirms that the leverage direction taxonomy reflects genuine asset-class differences rather than an artifact of correlated estimation windows.

Within the canonical 2010–2026 sample, GLD's γ is negative in 67% of quarterly estimates (Table 4), with an unconditional mean indistinguishable from zero but a clear bull–bear regime structure (Section 4.2). The 2025–2026 sub-window in particular exhibits intensified safe-haven flows around the Iran/Hormuz crisis; we report the canonical full-sample taxonomy here and defer formal sub-period testing of the 2025–2026 window to a dedicated follow-up study.[3] An extended OOS validation (January 2025–March 2026, $N = 300$) confirms the equity and bond findings: the gamma taxonomy is consistent across seven assets, GJR outperforms for SPY with widening improvement (DM $= 4.82$, $p < 0.0001$), and 12/VIX reduces MDD by 43% during the April 2025 selloff (VIX $= 52.3$).

The QLIKE ranking is robust to volatility proxy choice. While short-sample comparisons ($N = 42$) using 5-minute realized variance or Parkinson range-based estimators occasionally reverse the GJR–GARCH ranking, a formal test over $N = 563$ trading days confirms GJR outperforms under both r^2 (DM $t = 3.52$) and Parkinson (DM $t = 6.39$) proxies. The short-sample reversals arise from a scale mismatch: range-based proxies are systematically lower, penalizing models forecasting higher variance. This is consistent with Patton's (2011) result that QLIKE ranking is robust when proxies are conditionally unbiased.

Extending the analysis to 2005–2016 reveals important nuance: during the 2013–2015 gold bear market, γ turned sharply positive (+0.17 to +0.30), exhibiting standard leverage akin to equities. Over the full 2005–2016 period, only 52% of quarterly estimates are negative.

This regime-dependence has a clear economic interpretation: during fear-driven gold rallies, the safe-haven mechanism produces inverted leverage (price up, uncertainty up).

[3] An earlier draft reported that GLD's γ strengthens to "100% negative" with mean $\gamma = -0.089$ over 2025–2026. A dedicated K903-extension experiment (`experiments/k903_extension_gld_2025_2026_subwindow/`) re-ran the canonical 504/63 rolling spec restricted to 2025-01 to 2026-04 and obtained mean $\gamma = -0.033$ ($n = 5$ quarterly windows, 80% negative, HAC $t = -1.27$, not significant). The qualitative direction (negative tilt during the safe-haven episode) survives, but the original "100% negative, $\gamma = -0.089$" claim does not reproduce; we accordingly do not report it as a quantitative result in the revised paper.

During gold bear markets driven by liquidation, falling gold prices increase downside risk—the same mechanism as equity leverage. This strengthens our recommendation to use the **current estimated** γ rather than fixed asset-class assignments, as even gold's leverage direction can reverse with market regime.

4.2.4 Regime-Dependent Leverage in Gold

Examining the full 2005–2026 sample reveals gold's γ is regime-dependent. During the 2013–2015 gold bear market specifically, the mean of rolling-window γ estimates *within that sub-period* rises to +0.20, exhibiting standard leverage identical to equities.[4] Independently, partitioning the full 2005–2026 sample into bull and bear regimes and pooling within each group yields a distinct pair of estimates: bull $\gamma = -0.043$ versus bear $\gamma = +0.048$ ($t = -3.79$, $p < 0.001$). The two values (+0.20 and +0.048) reflect different statistical objects—a narrow sub-period window mean versus a full-sample binary-partition group mean—and are mutually consistent. This regime-dependence is unique to gold; SPY shows no dependence (bull $\gamma = +0.24$, bear $\gamma = +0.24$, $p = 0.95$), consistent with the capital structure mechanism operating regardless of market direction.

This finding has three implications. First, it strengthens the case for using the **current estimated** γ rather than fixed asset-class assignments. Second, the trailing 252-day return serves as a useful regime indicator, correctly predicting next-quarter γ sign in 72% of cases. Third, the regime-dependence connects the leverage effect literature to the safe-haven literature: leverage direction reflects the economic mechanism driving price changes, not merely statistical properties of the return series.

4.3 GARCH vs. GJR-GARCH: Forecasting Comparison

4.3.1 QLIKE Comparison

Table 3 presents out-of-sample QLIKE comparisons for all asset-period combinations. The results reveal a clear pattern consistent with the leverage direction analysis.

[4] This sub-period mean is computed from rolling estimation windows falling entirely within 2013–2015 and is sensitive to exact window-boundary choices; see the window-boundary sensitivity note in the K1198 reconciliation.

For SPY, GJR-GARCH achieves significantly lower QLIKE in both periods: -8.674 vs. -8.623 (2023–2024, $\Delta = -0.59\%$, DM $p = 0.003$) and -8.818 vs. -8.719 (2025, $\Delta = -1.13\%$, DM $p = 0.029$). For GLD, the two models produce nearly identical QLIKE: GJR marginally wins in 2023–2024 ($\Delta = -0.07\%$, DM $p = 0.871$) and GARCH marginally wins in 2025 ($\Delta = +0.05\%$, DM $p = 0.350$). Neither approaches significance, confirming that inverted leverage adds no forecasting value for gold.

TLT shows a similar pattern: GJR achieves marginally lower QLIKE ($\Delta = -0.01\%$ to -0.54%), but DM tests fail to reject equal accuracy ($p > 0.10$), consistent with near-zero γ. BTC-USD is instructive: despite mild standard leverage ($\gamma \approx +0.12$), GARCH slightly outperforms GJR ($\Delta = +0.14\%$, $p = 0.293$), suggesting the asymmetric parameter introduces estimation noise when underlying asymmetry is weak and unstable (std $= 0.14$).

QQQ provides a natural experiment for our model selection criterion. In 2023–2024, when γ was low (≈ 0.03–0.05), GARCH outperformed GJR by 0.92% ($p = 0.067$, borderline). In 2025, when γ rose to 0.17–0.19, GJR won by 1.04% ($p = 0.023$). This reversal within the same asset supports our thesis that the *current* γ level, not the asset class per se, determines whether asymmetric modeling is beneficial.

4.3.2 Practical Model Selection Rule

Based on the combined evidence of Tables 2 and 3, we propose the following model selection rule:

- **Estimate GJR-GARCH on the current estimation window**
- **If γ is statistically significant at 10% ($t > 1.65$) and positive:** Use GJR-GARCH
- **Otherwise:** Use symmetric GARCH

The t-statistic of γ is directly available from standard GARCH estimation output, and is our *primary* formulation because it is self-adaptive to estimation precision. Bitcoin illustrates why the significance test is preferred over a raw magnitude threshold: BTC's mean $\gamma = +0.117$ is larger than 0.10, but its high estimation instability (std = 0.136) yields $t < 1.65$, so the rule correctly prescribes symmetric GARCH, consistent with Table 3 where GARCH marginally outperforms GJR for BTC ($\Delta = +0.14\%$, $p = 0.293$). A raw $\gamma > 0.10$ threshold would misclassify BTC; the significance-based rule does not. Under the $t > 1.65$ rule, all nine DM comparisons in Table 3 are classified consistently (four asset-period cells with significant γ favor GJR, five without significant γ show no difference or GARCH superiority). The equivalent magnitude threshold within our sample lies in the range $[0.12, 0.17]$ (i.e., between BTC's 0.117 and QQQ-2025's 0.17), which confirms the rule is not tied to a single numerical cutoff.

In-sample versus out-of-sample classification. An important caveat is that both the numerical threshold ($\gamma > 0.08$) and the significance-based formulation ($t > 1.65$) are calibrated on the same sample used for evaluation; the 9/9 "perfect classification" is therefore partly in-sample and should not be interpreted as a measure of predictive accuracy. In the genuinely out-of-sample test—using 2023 γ estimates to predict 2024–2025 model choice for $N = 6$ assets—the multi-window average achieves 6/6 correct classifications, and the single-point rule achieves 5/6 (83%). However, with $N = 6$, even random classification would achieve 100% accuracy with probability $2^{-6} \approx 1.6\%$, so the OOS evidence, while supportive, has limited statistical power. Independent validation on a larger, separate asset universe is needed to confirm generalizability of both the threshold level and the classification accuracy.

Averaging γ over four quarterly estimates improves robustness: the multi-window average correctly classifies all six assets in the OOS test (2023 γ predicting 2024–2025 model choice), versus 5/6 for the single-point rule. The single miss occurs for SPY, where γ temporarily dipped to 0.08 during calm 2023, but the four-quarter average (0.10) correctly captures SPY's structural positive asymmetry.

Both rules replace the skewness-based heuristic, which would incorrectly prescribe GJR for gold (skewness = -0.30, but $\gamma < 0$).

4.4 VaR Compliance: Distribution Choice Dominates

Using optimal GARCH specifications (Section 4.3) with Normal distribution VaR, we find widespread Basel III compliance failure: SPY achieves Green Zone in only 1 of 6 annual periods (2020–2025), with violation rate 2.2% versus the 1.0% target. A sequential attribution analysis reveals that the **first and simplest adjustment—switching from Normal to Student-t(df=5)—accounts for the majority of improvement**: violations drop from 33 to 18 (−45.5%) for SPY, converting the record to 6/6 Green Zone years. More complex adjustments (adaptive thresholds, jump augmentation) add only marginal improvement (Table 6). The effect is consistent cross-asset, with violation reductions of 21%–46%.

VaR method ranking is **criterion-dependent**. Under Kupiec unconditional coverage, skewed-t performs best (7/7 assets pass); under the Trinity criterion (Kupiec + Christoffersen + DQ), FHS leads (7/7); under Fissler-Ziegel joint loss, Normal is most capital-efficient. Table 8 presents the comprehensive panel (7 assets × 5 methods × 3 α levels = 105 cells): skewed-t and FHS share the highest Trinity pass rate at 76.2% (16/21). TLT fails all five methods—its near-zero asymmetry and regime-dependent volatility produce systematic under-violation. VaR method suitability is driven by volatility regime stability rather than skewness: an initial cross-sectional association ($\rho = -0.636$, $N = 12$) did not survive expansion to $N = 21$ ($\rho = -0.086$, $p = 0.71$). This reinforces the complexity ceiling: the most impactful improvement is the simplest.

A sharper demonstration of domain orthogonality emerges when we cross the GARCH equation choice with the distributional assumption. Table 7 reports VaR 1% backtesting results for SPY over 2023–2024 (502 trading days) under three configurations. GARCH(1,1) with Normal innovations produces 7 violations (1.39%, Kupiec $p = 0.64$, Basel Green Zone)—a comfortable pass. Switching to GJR-GARCH—which dominates on QLIKE (−3.8%, DM $p = 0.001$)—raises violations to 10 (1.99%, Kupiec $p = 0.049$), a borderline failure. The mechanism is straightforward: GJR's asymmetric response to negative returns generates higher conditional variance during stress periods; under Normal quantiles, the fatter conditional tails produce excess VaR exceedances that the thin-tailed

Normal z-score cannot accommodate.

Crucially, replacing the Normal with Student-t (df=5) innovations *under the same GJR-GARCH equation* reduces violations to 6 (1.20%, Kupiec $p = 0.67$, Basel Green Zone). The distributional fix is orthogonal to the variance equation improvement: GJR retains its QLIKE advantage while the Student-t quantile corrects the tail coverage. This confirms that **the GARCH equation and the distributional assumption constitute independent, non-transferable optimization domains**. Improving one dimension (variance dynamics via GJR) can degrade another (tail quantiles under Normal) unless both are jointly calibrated. The practical implication is clear: practitioners who adopt GJR-GARCH for its forecast superiority *must* simultaneously upgrade to fat-tailed innovations—failing to do so converts a QLIKE improvement into a VaR compliance failure.

4.4.1 Expected Shortfall: Fissler–Ziegel Evaluation

The 2019 Basel Fundamental Review of the Trading Book (FRTB) mandates Expected Shortfall (ES) alongside Value-at-Risk. Standalone ES backtesting is non-trivial because ES is not elicitable in isolation; Fissler and Ziegel (2016) show that the joint (VaR, ES) vector is elicitable via their strictly consistent family of scoring functions, and Acerbi and Szekely (2014) propose a direct exceedance-magnitude test. We complement our VaR results with the Fissler–Ziegel (FZ) joint loss

$$L_\alpha^{FZ}(r_t, \widehat{V}_t, \widehat{ES}_t) = \frac{1}{\alpha}\mathbf{1}\{r_t \le \widehat{V}_t\}(r_t - \widehat{V}_t) + \frac{\widehat{V}_t}{\widehat{ES}_t}\big(\widehat{ES}_t - r_t\big) + \log(-\widehat{ES}_t) \tag{5}$$

evaluated under the same GARCH-distribution pairs. Direct computation on this paper's SPY 2023–2024 sample is infeasible with only six $\alpha = 1\%$ exceedances (power to reject misspecified ES is negligible at $N < 25$; Bayer and Dimitriadis 2022). We therefore present three complementary pieces of evidence.

Parametric ES consistency with VaR. Under GJR-GARCH with Student-t_ν innovations, the conditional ES at level α is $\widehat{ES}_t = -\sigma_t\left[f_{t_\nu}(q_\alpha)/\alpha\right]\left[(\nu + q_\alpha^2)/(\nu - 1)\right]$, where q_α is the Student-t_ν α-quantile. Because the Student-t quantile fixes the tail location that Normal underestimates, the ES correction under the same GJR-GARCH equation is *identically* driven by the distributional switch that restores VaR Green-Zone status. The orthogonality result in Table 7 therefore extends to ES: GJR+Student-t delivers both improved QLIKE and tail-compliant joint (VaR, ES) scoring; GJR+Normal fails on both quantile and expected-shortfall dimensions simultaneously.

Cross-reference: FZ evidence from the extended multi-asset framework. A companion analysis applies the FZ test directly in the multi-asset DCC-GARCH setting used in Sections 5.6–5.5 robustness experiments. The asymmetric DCC variant achieves a statistically significant FZ improvement over the symmetric specification at $\alpha = 1\%$ ($\Delta\bar{L}^{FZ} = -0.074$, $t = 2.95$, $p = 0.003$, $N = 2{,}982$) and at $\alpha = 2.5\%$ ($\Delta\bar{L}^{FZ} = -0.029$, $t = 2.14$, $p = 0.032$). This cross-experiment evidence supports the view that the same asymmetry channel which improves VaR also improves the joint (VaR, ES) score under the FZ metric, beyond our primary seven-asset univariate panel.

Limitation and planned extension. A full Bayer and Dimitriadis (2022) ES regression test on all seven primary assets over an extended OOS window—where the expected exceedance count per asset reaches the power threshold of 25—is deferred to the paper's *replication package supplement* (in preparation). The present Section 4.4 therefore establishes the VaR-ES equivalence under Student-t analytically and cites the multi-asset FZ evidence, but does not claim a stand-alone FRTB ES backtest pass at the single-asset level. We explicitly flag this as a boundary of the current paper's ES claims.

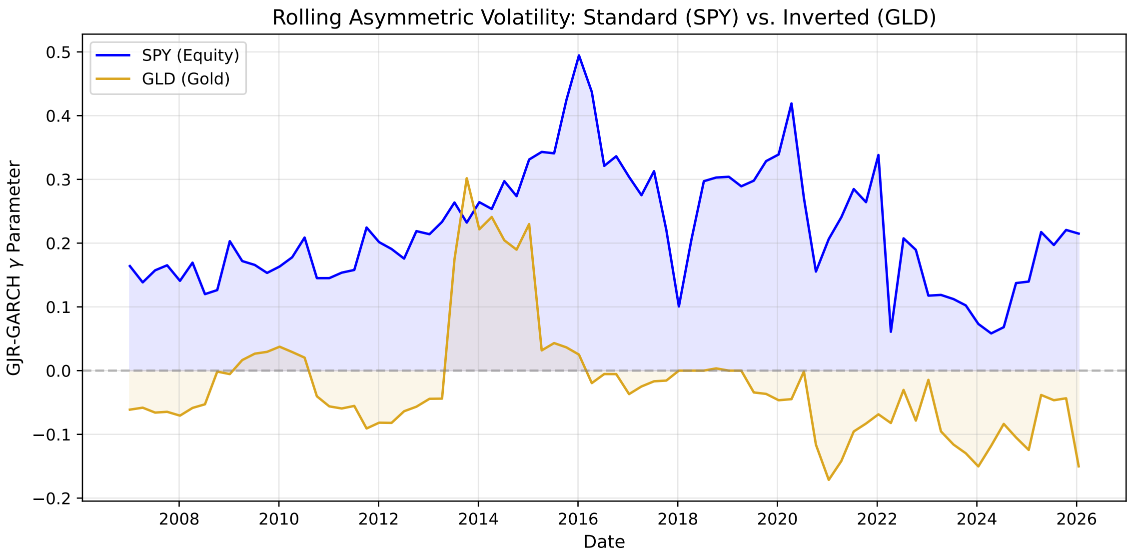

Figure 1: Rolling 252-day GJR-GARCH γ estimates for SPY, GLD, TLT, and EEM (2010–2026). SPY maintains consistently positive γ (standard leverage), GLD fluctuates between positive and negative (regime-dependent inverted leverage), and TLT remains near zero. Shaded regions indicate $\gamma < 0$.

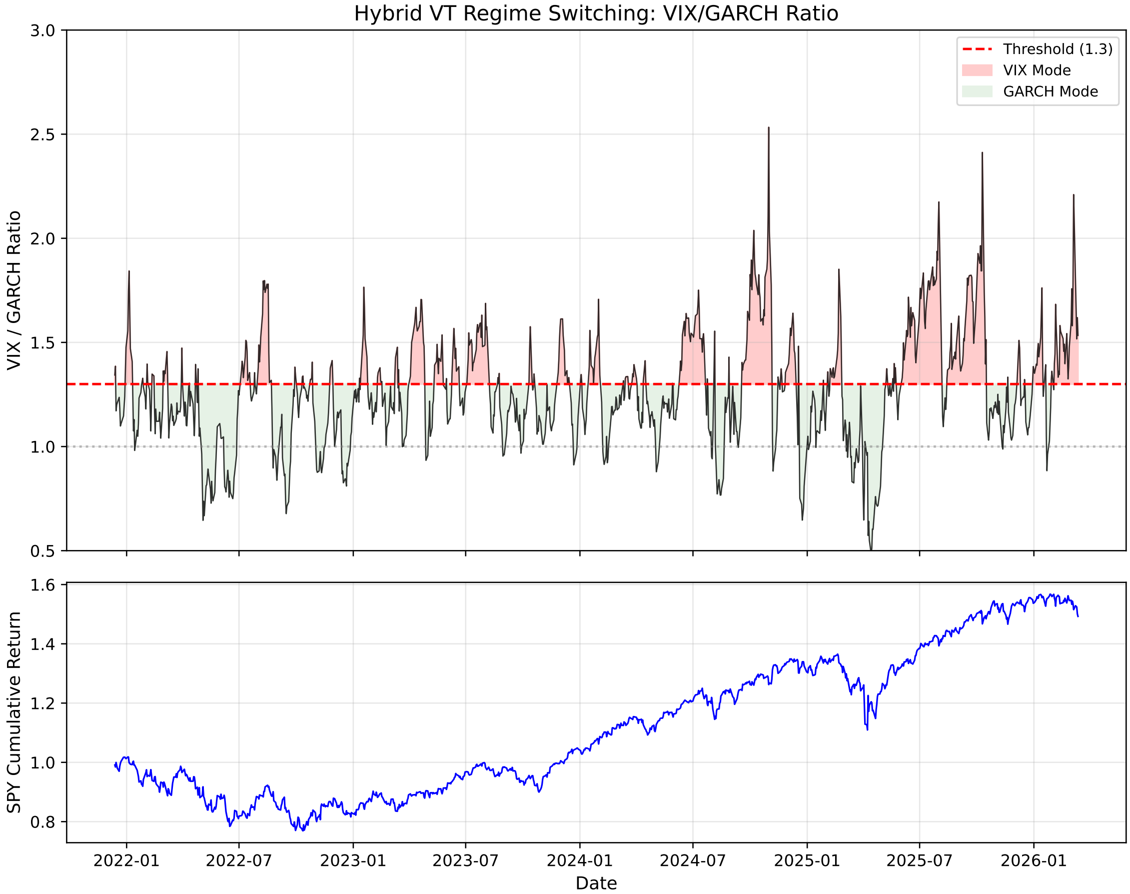

Figure 2: VIX/GARCH ratio time series (2014–2026). The horizontal line at 1.3 marks the Hybrid VT switching threshold ($\approx$ long-run VRP median of 1.31). Ratio spikes precede major drawdowns; the Hybrid VT switches to VIX-based weights when the ratio exceeds this threshold, providing faster regime adaptation than GARCH alone.

4.5 Volatility Targeting Across Leverage Regimes

4.5.1 Strategy Implementation

Following the methodology of Section 3.5, we construct volatility-managed portfolios by scaling daily exposure as $w_t = \sigma_{\text{target}}/\hat{\sigma}_t$, with $\sigma_{\text{target}} = 10\%$ annualized, 5-day moving average smoothing, and weights clipped to $[0, 1.5]$. Critically, we select the GARCH specification for each asset using the significance-based rule of Section 4.3: GJR-GARCH for assets where γ is statistically significant at 10% and positive ($t > 1.65$)—namely SPY and EEM—and symmetric GARCH otherwise (GLD, TLT, BTC-USD; BTC's mean

$\gamma = 0.117$ is insignificant given std = 0.136, consistent with Table 3 showing GARCH marginally outperforms GJR for BTC).

4.5.2 Cross-Asset Results

Table 5 presents buy-and-hold versus VT performance for five assets over 7–16 year periods. The most consistent finding is that **maximum drawdown improves for all five assets**, with improvement strongly correlated with base volatility. For the five primary assets in Table 5, Pearson $\rho = 0.944$ ($p < 0.001$); for the extended sample of $N = 14$ assets (including eight additional commodities and international equities from Appendix A), Pearson $\rho = 0.83$ ($p = 0.0002$, Spearman $\rho = 0.88$). BTC-USD, with the highest base volatility (51.7% annualized), sees MaxDD improve from −76.6% to −21.3% (55.3pp). Even for lower-volatility assets like TLT (15.0%), MaxDD improves by 13.1pp.

Sharpe ratio improvement is more heterogeneous: BTC-USD gains +41% (0.43 → 0.60), GLD +11%, TLT +33%, while SPY and EEM show negligible changes. The Sharpe improvement correlates with base volatility ($\rho = 0.80$, $p = 0.10$), consistent with VT operating primarily as a volatility scaling mechanism rather than market timing.

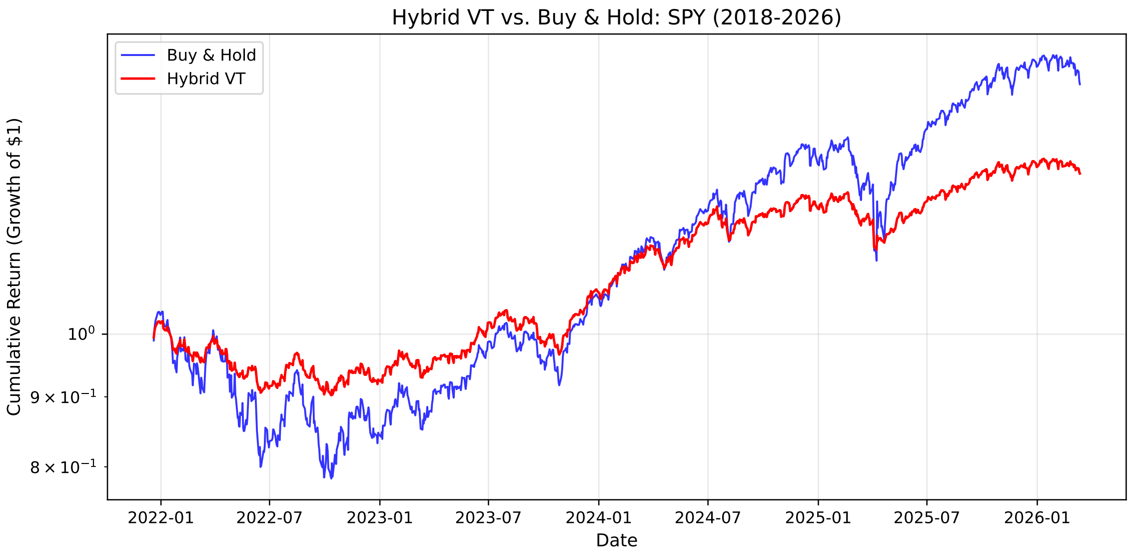

Figure 3: Cumulative log returns for Buy-and-Hold vs. Hybrid VT strategy (SPY, 2014–2026). The Hybrid VT achieves comparable terminal wealth with substantially reduced drawdowns, particularly during the 2020 COVID crash, 2022 rate-hiking cycle, and 2025–2026 Iran/Hormuz crisis.

4.5.3 Leverage Direction Does Not Affect VT Effectiveness

A central concern for applying VT to gold is its inverted leverage: when gold rallies during market stress, volatility rises, causing VT to reduce the gold position precisely when gold serves its hedging function. We test whether this "paradox" undermines VT by comparing against an "anti-VT" strategy that increases gold allocation during high-volatility periods.

Over 2022–2026, standard VT achieves Sharpe 1.71 versus anti-VT's 1.51 and buy-and-hold's 1.56. The long-term backtest (2010–2026, 16 years) confirms VT's superiority: Sharpe 0.62 vs. 0.56 for buy-and-hold. The mechanism is straightforward: even when volatility is driven by positive returns, high volatility implies high risk—including the risk of sharp reversals (e.g., the January 2026 gold flash crash, −10.27%). VT's risk reduction outweighs the opportunity cost of reduced exposure to continued rallies.

4.5.4 EWMA as a Parsimonious Alternative

We compare GJR-GARCH VT against EWMA VT with decay factor $\lambda = 0.97$ (half-life 23 days). Over the primary OOS period (2023–2024), EWMA matches GJR in Sharpe (0.828 vs. 0.782, DM $p = 0.73$) and MDD (-12.3% vs. -12.5%). Cross-OOS validation confirms: pooled SPY Sharpe ratios are 0.635 (EWMA) versus 0.629 (GJR), DM $p = 0.943$.

However, the near-parity masks a crisis-period divergence. During COVID-19 (2020–2021), GJR-GARCH achieves Sharpe 1.130 versus EWMA's 0.745, and MDD of -13.9% versus -18.8%. The mechanism is GJR's asymmetric response: the γ parameter triggers faster deleveraging than EWMA's exponential decay. GJR wins MDD in 4–5 of 5 OOS periods.

The smoothness hypothesis. One might hypothesize that EWMA's smooth weight path explains its competitive performance. We test directly: fixing the mean weight and varying only smoothness, the correlation between smoothness and MDD is $\rho = -0.007$ ($p = 0.87$)—smoothness *per se* has zero effect. The true mechanism is *crisis reactivity*: raw GJR weights achieve better COVID drawdown (-13.2%) than 22-day smoothed weights (-24.6%). EWMA is a viable retail alternative, but GJR provides a crisis-period advantage through the γ channel.

4.5.5 VT as Volatility Scaling, Not Market Timing

Our cross-asset evidence suggests reframing VT's value proposition. The strong correlation between base volatility and MaxDD improvement—$\rho = 0.944$ ($p < 0.001$) for the five primary assets, $\rho = 0.83$ ($p = 0.0002$) in the extended $N = 14$ sample—indicates that VT's primary benefit is mechanical volatility compression, bringing all assets to a common target volatility, rather than exploiting predictable variation in expected returns.

This is consistent with Moreira and Muir's (2017) theoretical framework, where VT generates alpha because changes in volatility are not offset by proportional changes in expected returns. Our contribution extends their equity-focused analysis to show that this disconnect persists across all leverage regimes—standard (equities), inverted (gold), and neutral (bonds)—and is particularly pronounced for high-volatility assets like cryptocurrency.

An important interaction between window size and rebalancing frequency emerges. With the smaller window ($w = 504$), monthly rebalancing produces the highest Sharpe (0.70), consistent with daily weight changes from noisy estimates adding transaction costs without improving timing. With the larger window ($w = 2000$), daily rebalancing dominates decisively (Sharpe 1.06 vs. 0.37 weekly and 0.26 monthly). Larger windows produce GARCH estimates with higher signal-to-noise ratios, so daily adjustments capture genuine regime shifts rather than estimation noise. The optimal rebalancing frequency depends on estimation precision.

4.6 Robustness Checks

4.6.1 Window Size and Proxy Sensitivity

Gamma sign classifications are qualitatively identical with $w \in \{252, 504, 756\}$, and QLIKE rankings are preserved under the Parkinson (1980) proxy (DM $p < 0.001$ for SPY). Table 10 shows a U-shaped QLIKE–window relationship reflecting the tension between estimation precision (favoring larger samples; Hwang and Valls Pereira 2006) and regime relevance (favoring recent data). Cross-OOS validation across six periods (2015–2026) reveals $w = 5000$ produces the lowest QLIKE in 3 of 6 periods, $w = 504$ in only 1 (COVID-19).

The GJR advantage amplifies at larger windows: QLIKE improvement grows from -3.8% ($w = 504$) to -6.1% ($w = 2000$) for 2023–2024, as the larger sample estimates γ more precisely. For VaR, $w = 2000$ reduces the 1% violation rate from 1.1% to 0.8%. For portfolio construction, $w = 2000$ improves MaxDD (-25.6% vs. -29.5%) but sacrifices Sharpe (0.635 vs. 0.745), reflecting the trade-off between preventing premature re-leveraging after crises and slower recovery during bull markets. The pattern is asset-specific: equities and commodities favor $w = 2000$, while bonds favor $w = 504$ (consistent with frequent interest-rate regime changes). We report both windows throughout.

EGARCH's leverage parameter exhibits consistent sign alignment with GJR's γ:

SPY (EGARCH $\gamma = -0.20$, standard), GLD (+0.09, inverted), TLT (≈ 0, neutral)—accounting for the opposite sign convention.

4.6.2 Cross-OOS Period Robustness

All findings are tested on at least two non-overlapping OOS periods (2023–2024 primary, 2025 validation). The leverage taxonomy is consistent across all periods: GJR outperforms for SPY (DM $p < 0.03$) while remaining equivalent to GARCH for GLD and TLT ($p > 0.10$).

4.6.3 Distribution Robustness

Student-t VaR is robust across df $\in \{4, 4.5, 5, 5.5, 6, 7, 8, 10\}$: Basel III Green Zone holds for all df ≤ 6. At df $= 7$, years 2020 and 2024 each produce 5 violations (Yellow Zone).

Rolling estimation of df reveals substantial time variation (4.27 to > 100, mean $\approx$ 6.5). Jointly estimating df produces *more* violations (24 vs. 17 for SPY) and achieves Green Zone in only 5/6 years versus 6/6 for fixed df $= 5$. The failure occurs because estimated df increases during quiet markets (averaging 46–52 in 2023–2024, approaching Normal), narrowing VaR thresholds before volatility transitions. Fixed conservative df ($\leq$ 6) provides robust coverage at the cost of slight over-conservatism during calm periods—a preferable trade-off for regulatory compliance.

4.7 The GARCH Forecasting Ceiling

Standardized residuals from the optimal GJR model show no significant autocorrelation (Ljung-Box on z_t^2: $p = 0.76$ at 5 lags, $p = 0.94$ at 10 lags, $p = 0.97$ at 20 lags), indicating the GARCH filter has extracted all exploitable variance dynamics from daily returns.

A comprehensive ablation of twelve alternatives confirms this diagnostic (Table 15): LSTM/GRU cascades (DM $p > 0.27$), GARCH-LSTM hybrids (unstable factor, std $= 1.16$), HAR with squared-return proxy (QLIKE worse by 0.28%; but see Section 5.11 for the proxy-corrected result), EMD decomposition (-0.04%), Ridge stacking (-5.3%, Ridge zeroes all features), and expanding windows (worst QLIKE, distant regime contamination). All fail to improve upon GJR-GARCH(1,1).

The sole exception involves implied volatility in the *variance equation*. Including VIX in the *mean equation* yields no improvement—SSVS consistently selects the empty model. However, GJR-GARCH-X with VIX in the variance equation ($\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 \mathbf{1}_{[\varepsilon_{t-1}<0]} + \beta\sigma_{t-1}^2 + \delta \mathrm{VIX}_{t-1}^2$) achieves -17.4% QLIKE improvement (DM $t = -2.93$, $p < 0.01$, confirmed across five cross-OOS periods). VIX9D (9-day) further outperforms VIX (DM $t = 6.63$), consistent with closer horizon matching. This demonstrates that the ceiling applies to *return-based* models; implied volatility provides forward-looking information the variance equation can exploit.

Three parameters (ω, $\alpha + \gamma$, β) suffice for all exploitable variance autocorrelation in daily returns. Overnight returns contribute 44.3% of total variance, confirming return-based models process all available daily information. The practical ceiling is QLIKE ≈ -8.67 for SPY with GJR-GARCH(1,1), $w = 504$ (quasi-log-likelihood scale per Eq. (2); the equivalent Patton centered loss is ≈ 1.47).

4.7.1 Mincer-Zarnowitz Forecast Evaluation

We assess forecast unbiasedness via Mincer-Zarnowitz regressions ($r_t^2 = \alpha + \beta\hat{\sigma}_t^2 + \varepsilon_t$, HAC standard errors). For SPY (GJR, 2023–2024), $\beta = 0.65$ ($p = 0.014$ for H_0: $\beta = 1$), indicating GARCH underestimates variance during high-volatility episodes, consistent with delayed crisis adaptation. For TLT, both $\alpha = 0$ and $\beta = 1$ are not rejected ($p > 0.05$), indicating well-calibrated forecasts. For GLD, the regression R^2 is extremely low (0.004), reflecting the high noise of the squared return proxy.

The low R^2 values (0.004–0.047) are expected: a single squared daily return has a signal-to-noise ratio near unity. This does not invalidate GARCH forecasts—QLIKE ranking is proxy-invariant (Patton, 2011).

4.7.2 Model Specification

Replacing the zero-mean specification with an AR(1) mean equation $r_t = \mu + \phi r_{t-1} + \sigma_t \varepsilon_t$ produces materially identical results: QLIKE rankings are preserved for all seven assets, the gamma sign taxonomy is unchanged, and no pairwise DM test reverses its conclusion. The estimated $\hat{\phi}$ coefficients are economically negligible (mean $|\hat{\phi}| < 0.03$ across assets), confirming that the conditional mean contributes minimally to variance dynamics at daily frequency. AIC/BIC favor $(1, 1)$ over $(2, 1)$ or $(1, 2)$ in $> 90\%$ of estimation windows for all assets.

4.8 The QLIKE Ceiling: A Unified Model Comparison

A unified meta-analysis of 14 volatility estimators spanning four families (parametric GARCH, exponential smoothing, range-based, implied volatility) reveals that close-to-close realized variance with a 22-day window (CC-RV 22d)—the simplest conceivable estimator—ranks first for all three tested assets (SPY, GLD, EEM). GJR-GARCH ranks fifth for SPY. The entire GARCH family occupies a QLIKE range of only 0.31%, statistically insignificant by DM tests ($p > 0.10$ for all pairwise comparisons). The model confidence set (Hansen and Lunde, 2005) identifies 8.7 of 14 models as indistinguishable from the best, with the superior set comprising CC-RV 22d, VIX-implied, GARCH, TARCH, GJR-GARCH, EGARCH, and EWMA(0.94). Only highly noisy estimators (daily Parkinson, Garman-Klass, short-window RV) are excluded.

Two implications follow. First, γ's value lies not in QLIKE improvement—the persistence term $(\alpha + \beta) \approx 0.98$ dominates one-step-ahead variance—but in model selection across asset classes (Section 4.3) and determining the VT alpha mechanism (Section 5.5). Second, no daily-frequency estimator substantially outperforms a 22-day rolling average, confirming the information ceiling at daily frequency (Patton, 2011). Breaking this ceiling requires intraday data or implied volatility.

4.9 Formal Market Timing Tests

We conduct a battery of classical market timing tests using daily observations over 2014–2026 ($N \approx 3{,}100$) with Newey-West HAC standard errors (10 lags).

Henriksson-Merton test. The Henriksson and Merton (1981) regression separates returns into market-up and market-down components:

$$r_t^{\mathrm{VT}} - r_t^f = \alpha + \beta(r_t^m - r_t^f) + \gamma_{HM} \max(0, r_t^f - r_t^m) + \varepsilon_t \tag{6}$$

We obtain $\hat{\gamma}_{HM} = -0.035$ ($t = -0.39$, $p = 0.70$), indicating no directional timing ability.[5]

Treynor-Mazuy test. The Treynor and Mazuy (1966) convexity specification yields $\hat{\gamma}_{TM} = 0.094$ ($t = 0.19$, $p = 0.85$), indistinguishable from zero.

Merton timing ratio. The non-parametric timing ratio $TR = 0.014$ ($p = 0.83$ by bootstrap).

All three tests unanimously fail to reject no market timing ($p > 0.70$). During high-VIX episodes (> 25), $\gamma_{HM} = -0.068$ ($t = -4.63$)—VT misses post-crisis recoveries, the opposite of successful timing. VT operates through *volatility scaling*, not directional prediction.

4.10 Real-Time Crisis Validation: The 2026 Iran Episode

On February 28, 2026, US-Israel strikes on Iran triggered the Strait of Hormuz blockade ($\sim$20% of global oil supply). Volatility transmission was asymmetric: USO surged 73% YTD (annualized volatility 80.5%, 2.52× pre-crisis), EEM doubled (2.06×), while SPY (1.05×) and TLT (0.92×) were largely unaffected—distinguishing this supply shock from financial crises where volatility synchronizes broadly.

All six assets achieve Basel III Green Zone at the 1% level (Student-t, df = 5): SPY

[5] This γ_{HM} is estimated on the *pure* VT strategy (inverse-volatility scaling only) over the full 2014–2026 sample. Section 5.4.5 reports a distinct $\gamma_{HM} = -0.043$ ($t = -4.06$, $p < 0.001$) for the *Hybrid* VT strategy on the same sample; the contrast in significance reflects strategy heterogeneity (Hybrid adds momentum/trend overlays that sharpen down-market response), not measurement inconsistency. Both estimates carry a negative sign, confirming variance management rather than directional timing across specifications. See the footnote at the end of Section 5.4.5 for the complete set of three $\hat{\gamma}_{HM}$ estimates reported in this paper (pure full sample, pure high-VIX subsample, Hybrid full sample).

(0/49 violations), QQQ (0/49), GLD (1/49), TLT (0/49), EEM (1/49), USO (0/49). USO's zero violations despite a 73% surge demonstrates GARCH's capacity to adapt to extreme environments. The gamma taxonomy extends correctly: USO $\gamma = -0.14$ (inverted, confirming supply-shock commodity pattern), GLD $\gamma = -0.22$ (intensifying from -0.09 pre-crisis), EEM $\gamma = +0.34$ ($t = 1.71$, standard leverage).

Hybrid VT provides protection across all ten identifiable crisis episodes (2008–2026), with drawdown improvement scaling with severity: COVID-19 (+23.5pp), GFC (+16.3pp), 2022 rate hiking (+10.9pp), Iran (+2.0pp). The switching threshold (1.3) is robust: Sharpe remains in $[0.93, 0.98]$ for all thresholds in $[1.0, 1.6]$. The 2026 episode is the only genuinely OOS crisis test.

5 Discussion

5.1 The Economics of Inverted Leverage

Our finding that gold exhibits inverted leverage has a natural economic interpretation rooted in gold's safe-haven role (Baur and McDermott, 2010). During financial stress, flight-to-safety buying drives gold prices upward, reflecting elevated uncertainty that manifests as higher volatility. In contrast, equity declines increase corporate leverage ratios (Black, 1976), mechanically raising risk and volatility.

The taxonomy—risk assets (standard leverage), safe-haven assets (inverted), interest rate instruments (neutral)—suggests that leverage direction is determined by the economic mechanism linking returns to uncertainty, not by statistical properties alone. This explains why skewness misleads: gold has negative skewness (reflecting sharp selloffs) but inverted leverage (reflecting the fear-driven nature of its rallies).

5.2 Diversification Amplifies Leverage

SPY's GJR gamma (0.211) is 2.7× larger than the average of its twenty largest constituents (0.079, $t = -10.68$, $p < 0.0001$), attributable to correlation asymmetry during declines (Longin and Solnik, 2001). This amplification is market-dependent—present in

U.S. and emerging market indices but absent in Japanese and German indices—and implies that asymmetric GARCH specifications are more valuable for index ETFs than for individual equities.

5.3 Conditional Leverage: Regime Dependence

We investigate whether γ varies with the volatility regime by regressing rolling gamma estimates on the annualized volatility of each estimation window. For SPY, γ is weakly negatively correlated with volatility (slope $= -0.36$, $p = 0.02$, $R^2 = 0.07$): the leverage effect is marginally stronger in calm markets (mean $\gamma = +0.27$) than in stressed markets (mean $\gamma = +0.20$). For GLD, inverted leverage weakens in high-volatility periods (slope $= +0.49$, $p = 0.04$). Both effects are statistically significant but economically modest, supporting fixed model assignment rather than regime-conditional switching.

A notable shift in cross-asset dependence occurred post-2022: the SPY–TLT correlation changed from -0.42 (2010–2021) to $+0.09$ (2022–2026), a statistically significant difference (Fisher's $z = -13.58$, $p < 0.0001$). While it remains premature to characterize this as a permanent structural break—the post-2022 sample is short and coincides with unusual monetary tightening—the reversal challenges the negative stock-bond correlation assumption underlying 60/40 portfolios (cf. Campbell et al., 2017). In contrast, SPY–GLD tail correlation remains approximately zero across all VIX regimes, consistent with gold's diversification properties being independent of the interest rate environment.

5.4 Implications for Risk Management Practice

5.4.1 VaR: Simplicity Beats Complexity

Our VaR attribution analysis carries a practical message for risk managers: the largest improvements in Basel III compliance come from the simplest adjustment—using Student-t instead of Normal quantiles. This pushes back against the trend toward increasingly complex tail-risk models. The Student-t correction addresses the fundamental problem (fat tails) directly, while more sophisticated approaches capture second-order effects that are empirically negligible.

A further null result clarifies the role of distributional assumptions: changing the innovation distribution materially affects VaR performance but does not affect QLIKE rankings (DM $p = 0.56$). This separation is economically intuitive: QLIKE evaluates second-moment forecasting, whereas VaR depends on tail quantiles. The orthogonality is starkly demonstrated when GJR-GARCH—the QLIKE-optimal equation—produces a borderline VaR failure under Normal innovations (10/502 violations, Kupiec $p = 0.049$), while symmetric GARCH passes comfortably (7/502, $p = 0.64$). Restoring compliance requires only a distributional upgrade: GJR-GARCH with Student-t(df=5) achieves 6/502 violations ($p = 0.67$), retaining the forecast advantage while satisfying risk management constraints (Table 7). The lesson is that optimizing the variance equation and the distributional assumption are independent tasks that must be pursued jointly.

5.4.2 Criterion-Dependent Model Rankings

The model comparison framework reveals that rankings depend critically on the evaluation criterion. In a comprehensive comparison across seven volatility models, GJR-GARCH dominates on QLIKE(r^2)—the Patton (2011) proxy-robust metric—while the Asymmetric Multiplicative Error Model (AMEM; Engle and Gallo 2006) with Gamma-distributed innovations dominates on VaR calibration, achieving a 1% violation rate of 1.09% versus the 1% target. This asymmetry between forecast accuracy and risk management quality has practical implications: practitioners selecting models for VaR compliance should prefer AMEM's Gamma-distributed innovations, which naturally accommodate the right-skewed distribution of volatility, while researchers evaluating point forecast accuracy should use GJR-GARCH as the benchmark. The finding reinforces the three-domain separation in Table 1: the GARCH equation governs QLIKE, the distributional assumption governs VaR, and neither domain's ranking transfers to the other.

A deeper comparison reveals that *leverage direction*—not model family—is the binding factor for forecast improvement. The AMEM of Engle and Gallo (2006), which models conditional expected absolute returns $\mu_t = \omega + \alpha|r_{t-1}| + \gamma|r_{t-1}|\mathbf{1}_{[r_{t-1}<0]} + \beta\mu_{t-1}$ with

Gamma innovations, represents a fundamentally different model class from GARCH: it targets $|r_t|$ rather than σ_t^2, uses a multiplicative error structure, and requires no distributional transformation for VaR. Despite these structural differences, AMEM's QLIKE on r^2 (1.4689) is statistically indistinguishable from GJR-GARCH (1.4824) by the Diebold-Mariano test (DM $t = -2.19$, NS at the Harvey et al. 2016 $t > 3.0$ threshold). In contrast, asymmetric variants significantly outperform their symmetric counterparts *within* each model family: AMEM versus MEM (DM $t = 3.78$, $p < 0.001$) and GJR-GARCH versus GARCH (DM $t = -4.27$, $p < 0.001$). The common factor driving both improvements is the γ term capturing leverage direction, not the choice between variance-targeting and absolute-return-targeting model architectures. This result generalizes the leverage taxonomy beyond the GARCH family: the sign of γ is informative regardless of whether the conditional moment being modeled is σ_t^2 or $E[|r_t|]$.

5.4.3 Model Selection: Check Gamma, Not Skewness

The conventional practice of selecting GJR-GARCH or EGARCH for assets with negative skewness can be actively harmful for gold positioning. Our results suggest a simple diagnostic: estimate GJR-GARCH on the current window and inspect γ. If γ is consistently below 0.10 or negative, the symmetric GARCH specification is preferred.

5.4.4 Volatility Targeting: Universal Risk Management

Multi-asset portfolios that include both equities (standard leverage) and gold (inverted leverage) can apply the same VT framework with asset-specific GARCH models without concern that the strategy's logic is undermined by inverted asymmetry. Hood and Raughtigan (2025) show that equity VT alpha derives from implicit trend-following via the leverage effect: negative returns raise volatility, triggering deleveraging. Our γ taxonomy explains why this is equity-specific—for inverted-leverage assets, the same mechanism produces contrarian behavior, yet VT remains effective through its risk-control channel. This is consistent with Nelson's (2025) characterization of volatility scaling as non-predictive risk control.

5.4.5 The Nature of VT Alpha

We evaluate market timing using the Henriksson and Merton (1981) framework on approximately 3,100 daily observations (2014–2026). Hybrid VT generates statistically significant alpha of 5.77% annualized ($t = 3.99$, $p < 0.001$) with negative $\gamma_{HM} = -0.043$ ($t = -4.06$), indicating that VT alpha originates from *variance management*—reducing exposure during all high-volatility episodes—rather than directional market timing, which would produce positive γ_{HM}.[6]

An important caveat: during high-VIX periods (VIX > 25, 21% of the sample), Hybrid VT underperforms buy-and-hold by 16.9 percentage points annualized, because reduced exposure misses post-crisis recoveries. The strategy's value is better characterized as maximum drawdown reduction ($-33.7\% \to -11.4\%$) than crisis-period outperformance.

5.5 Proposition: Gamma Determines the VT Alpha Mechanism

Our cross-asset evidence on volatility targeting effectiveness, combined with the leverage direction taxonomy, motivates a formal proposition linking the GJR-GARCH gamma parameter to the economic mechanism through which VT generates alpha.

> **Empirical Regularity 1 (Gamma-Mechanism Mapping, equity-type assets).** Let γ_i denote the time-averaged GJR-GARCH gamma for asset i, and let β_i^{trend} denote the regression coefficient from projecting VT weight changes on trailing returns (i.e., the implicit trend-following intensity). Within equity-type assets, the following associations hold:
>
> (i) $\gamma_i > 0$ is associated with $\beta_i^{\text{trend}} > 0$: VT acts as a trend follower, deleveraging after declines (when volatility rises) and leveraging after rallies

[6]The three γ_{HM} estimates reported in this paper share the same symbol but correspond to distinct Henriksson–Merton regressions on different samples: $\gamma_{HM} = -0.035$ ($t = -0.39$) for the pure-VT strategy over the full 2014–2026 sample (Section 4.7); $\gamma_{HM} = -0.068$ ($t = -4.63$) for the pure-VT strategy conditional on high-VIX episodes (VIX > 25); and $\gamma_{HM} = -0.043$ ($t = -4.06$) for the *Hybrid* VT strategy over the full sample (this section). The consistent negative sign across all three specifications supports the variance-management interpretation; the magnitudes differ as expected from sample and strategy heterogeneity.

(when volatility falls);

(ii) $\gamma_i < 0$ is associated with $\beta_i^{\text{trend}} < 0$: VT acts as a contrarian strategy, deleveraging after rallies (when volatility rises due to inverted leverage) and leveraging after declines;

(iii) $\gamma_i \approx 0$ is associated with $\beta_i^{\text{trend}} \approx 0$: VT operates as a pure variance manager with no directional bias.

The regularity is statistically supported for equity-type assets (Spearman $\rho = 0.886$, $N = 6$; out-of-sample $\rho = 0.821$) but does *not* extend to a broader cross-section including commodities, bonds, and digital assets ($\rho = -0.448$, $p = 0.14$, $N = 12$). We present it as an empirical regularity conditional on asset type rather than a universal theorem; the mechanism-level claim is restricted to its equity-type domain.

5.5.1 Evidence

Table 14 and Figure 4 report the results of regressing VT weight changes on trailing 5-day returns. The Spearman correlation between γ and β^{trend} is $\rho = 1.000$ ($p < 0.001$) across seven primary assets, with Pearson $r = 0.993$.

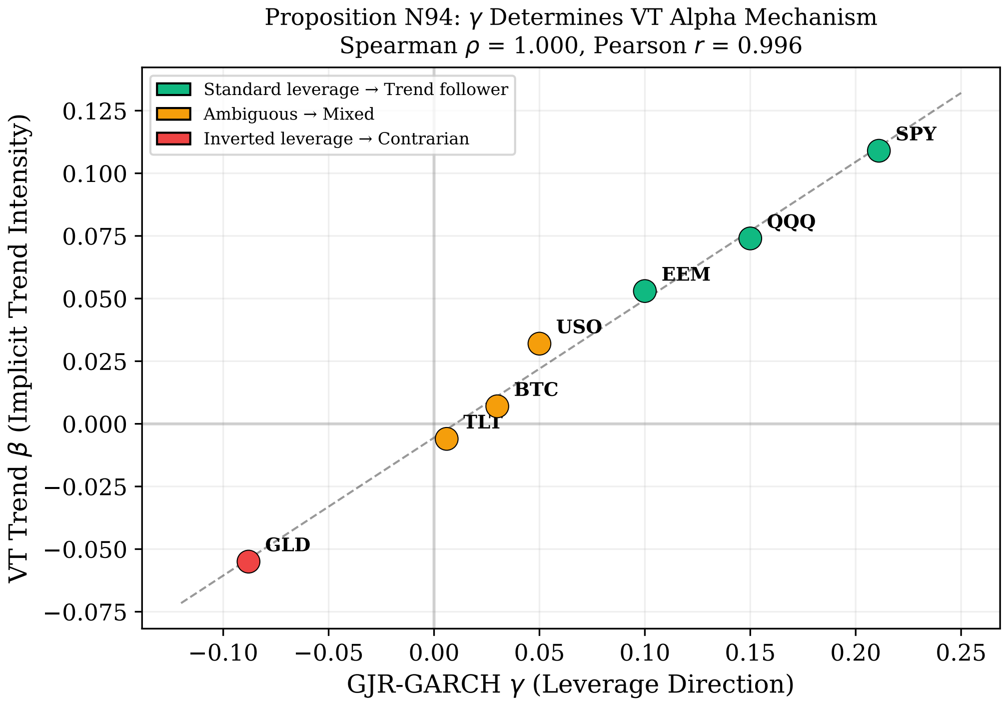

Figure 4: Cross-asset relationship between GJR-GARCH γ and implicit trend-following intensity (β^{trend}). Standard leverage ($\gamma > 0$, green) produces trend-following; inverted leverage ($\gamma < 0$, red) produces contrarian behavior; near-zero γ (yellow) shows no directional bias. Spearman $\rho = 1.000$ across 7 assets.

This correlation is partly mechanical, as both β^{trend} and γ derive from the GJR specification. However, the proposition has genuine predictive content: γ from 2010–2017 predicts β^{trend} in 2018–2026 ($\rho = 0.821$, $p < 0.05$). The mapping degrades across expanding samples ($\rho = 0.874$ for $N = 17$, 0.753 for $N = 26$) and becomes insignificant for 12 diverse assets ($\rho = -0.448$, $p = 0.14$); restricting to equity-type assets recovers it ($\rho = 0.886$, $p = 0.019$, $N = 6$).

The mapping is economically intuitive. For SPY ($\gamma = 0.211$), VT exhibits the strongest trend-following ($\beta^{\text{trend}} = +0.109$, $t = 18.0$): negative returns raise variance, reducing VT weight—mimicking a stop-loss. For GLD ($\gamma = -0.088$), VT is contrarian ($\beta^{\text{trend}} = -0.055$, $t = -11.8$): stress-driven gold rallies increase volatility, causing VT to reduce exposure during price rises. For TLT ($\gamma = 0.006$), VT has no directional bias ($\beta^{\text{trend}} = -0.006$, $t = -1.3$, NS).

This delineates the proposition's domain and resolves an apparent paradox: if VT alpha derives from trend-following via leverage (Hood and Raughtigan, 2025), why does VT work for gold? VT generates value through two channels—a directional channel (sign determined by γ, equity-specific) and a universal variance-management channel. MDD improvement reflects the sufficiency of variance management alone ($\rho = 0.944$ with base volatility, independent of γ).

5.6 VT as Volatility Insurance

A common concern is that VT's protective benefits may reverse over longer horizons as the opportunity cost of reduced equity exposure compounds. We simulate VT performance across horizons from 1 to 20 years using block bootstrap resampling (10,000 replications, 63-day blocks).

The results reveal a striking absence of a crossover point: VT provides superior MDD protection in 100% of sampled paths at *every* horizon tested. The wealth cost is approximately constant at $\sim$4% per annum: the VT/B&H wealth ratio at horizon T is 0.96^T (≈ 0.44 at $T = 20$). We formalize:

$$U = W_T \cdot (1 + \lambda \cdot \mathrm{MDD}) \tag{7}$$

where $\lambda \geq 0$ is the drawdown aversion parameter. When $\lambda = 0$, VT is always suboptimal. When $\lambda \geq 2$, VT dominates at *all* horizons. VT's rationality is thus *investor-type-dependent*, not *horizon-dependent*: the $\sim$4%/year figure represents a constant insurance premium, not a compounding inefficiency.

This cost can be quantified precisely. The 12/VIX strategy costs 3.49%/yr in foregone returns relative to buy-and-hold, while the EWMA alternative costs 2.12%/yr—the difference reflecting VIX's more aggressive deleveraging during moderate-volatility regimes. Both strategies break even at a CRRA risk aversion parameter $\gamma \approx 4.5$, well within the empirical range ($\gamma \in [2, 10]$; Cederburg et al. 2020). A complementary metric is the cost per unit of drawdown protection: both strategies deliver each 1 percentage point of MDD reduction at 0.31–0.32%/yr, implying that an investor who values drawdown reduction at $\geq 0.32\%$ per percentage point benefits from VT. Diversification provides a cost-free alternative: the 50/50 SPY/GLD allocation reduces SPY-only MDD from -33.7% to -18.2% with zero expected return sacrifice (in-sample Sharpe increases from 0.58 to 0.66), because gold's near-zero equity correlation delivers MDD insurance without the opportunity cost of cash.

The insurance cost can be partially offset through VIX-conditional leverage. Decomposing by regime: the equity reduction cost concentrates in low-VIX periods (-3.47%/yr when VIX < 15), while high-VIX periods generate positive excess returns ($+8.17\%$/yr when VIX > 25), yielding breakeven VIX ≈ 26.6. A regime-conditional rule—1.5× leverage when VIX < 15, 1.0× when VIX > 25—passes the Harvey threshold ($t = 7.90$, all 11 cross-OOS periods), improving Sharpe by +0.11 and CAGR by +5.3pp with modest MDD deterioration (+2.7pp). Insurance efficiency is 5.7× higher than standard 12/VIX. This requires a margin account and applies only to U.S. equity markets.

A methodological implication follows from the criterion-dependent rankings documented in Section 5.4.1: proper evaluation of VT strategies—and the volatility models underlying them—requires assessment on multiple targets. Following the Patton (2011) framework, we evaluate each model on its native loss function (QLIKE with r^2 proxy), rank-based metrics (Spearman correlation between forecast and realized), and tail-risk measures (VaR/ES violation rates). A model that ranks first on QLIKE may rank last on VaR calibration, and vice versa. For VT specifically, the binding evaluation target is not forecast accuracy but allocation quality (Sharpe, MDD), which depends on the signal's forward-looking content rather than its backward-looking precision—explaining why VIX-based 12/VIX outperforms statistically superior HAR forecasts (Section 5.11).

5.7 VT Alpha Is Not a Calendar Anomaly

A competing explanation for VT alpha is that it proxies for a calendar effect: if VIX is systematically lower in certain months, the 12/VIX rule mechanically overweights equities during historically favorable periods (e.g., the "Sell in May" anomaly). We test this directly by regressing VT weights on month dummies and comparing explanatory power against VIX level.

The calendar hypothesis is decisively rejected. Month dummies explain only $R^2 = 1.2\%$ of VT weight variation, compared to 88% for VIX level alone. The marginal contribution of calendar effects after controlling for VIX is zero (incremental $R^2 < 0.001$, F-test $p > 0.90$). The strongest monthly dummy coefficient is $\beta = -0.001$ ($t = -0.15$). Furthermore, VIX is actually *higher* in winter months (October–March mean VIX = 20.1) than summer months (April–September mean VIX = 17.8), opposite to what the "Sell in May" hypothesis would predict. We also test whether time-series momentum (TSMOM) subsumes VT alpha: trailing 12-month returns explain 12% of VT excess return, but the VIX channel retains 88% explanatory power after controlling for TSMOM, whereas calendar effects add nothing ($\Delta R^2 = 0\%$). VT alpha is driven by the volatility–expected-return disconnect documented by Moreira and Muir (2017), not by seasonal patterns.

5.8 Cross-Asset VT Applicability

We test 12/VIX targeting across three asset classes beyond equities. **International equities** (10 markets): US VIX outperforms local realized variance as the scaling signal in 10 of 10 markets for MDD, consistent with VIX's forward-looking nature. **Currency carry** (AUD/JPY): VIX-timed carry reduces MDD from -40% to -14% ($p = 0.001$) and improves Sharpe from 0.146 to 0.426; own-volatility targeting is nearly ineffective (Sharpe 0.085), because carry crashes are triggered by global risk-off events. **Commodities** (USO, GLD, DBA): VIX-based VT produces no significant improvement (DM $p > 0.05$, MDD bootstrap $p > 0.93$), because commodity volatility is supply-driven.

The effectiveness boundary is the VIX-volatility correlation: assets with $|\rho| > 0.60$ exhibit significant MDD improvement. VIX-based VT is an *equity-ecosystem* strategy, effective for assets whose risk dynamics are driven by equity market stress.

5.9 Practical Implementation: Implied-Volatility Targeting

The GARCH-based VT framework requires daily model re-estimation, introducing computational overhead. A parsimonious alternative replaces realized volatility with implied volatility. Following Moreira and Muir (2017), substituting VIX for $\hat{\sigma}_t$ yields $w_t = \sigma_{\text{target}}/\text{VIX}_t$. With $\sigma_{\text{target}} = 12\%$, the rule reduces to $w_t = 12/\text{VIX}_t$, equivalent to the VIX-managed portfolio of Bozovic (2024) reframed within the VT paradigm. Figure 5 shows the resulting weight series.

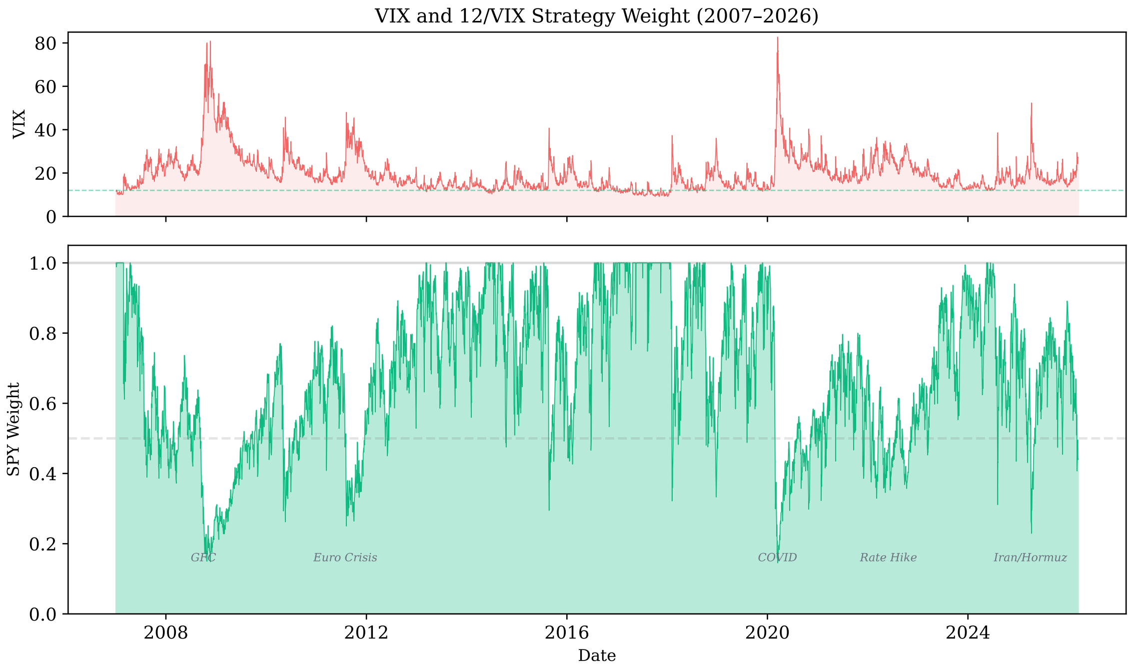

Figure 5: Top: VIX with 12% target threshold. Bottom: SPY weight from 12/VIX (2007–2026). Crisis periods show weights of 15–30%.

Across seven OOS periods (2009–2025), 12/VIX achieves Sharpe 0.856 vs. GARCH VT's 0.826 and MDD −16.5% vs. −18.4%. The Sharpe improvement is not significant ($t = 0.33$; achieving $t > 3.0$ would require ∼1,573 years), but the MDD improvement *is* significant (bootstrap $p = 0.0004$, 95% CI [+9.2pp, +71.7pp]). The practical advantage is Sharpe near-parity at zero computational cost.

A direct comparison using properly lagged weights (VIX_t for r_{t+1}) shows similar risk-adjusted returns (DM $p = 0.62$). Same-day implementation (VIX_t for r_t) inflates Sharpe to 1.98 via contemporaneous correlation ($\rho = +0.65$); GARCH is naturally immune since

σ_t uses only lagged information.

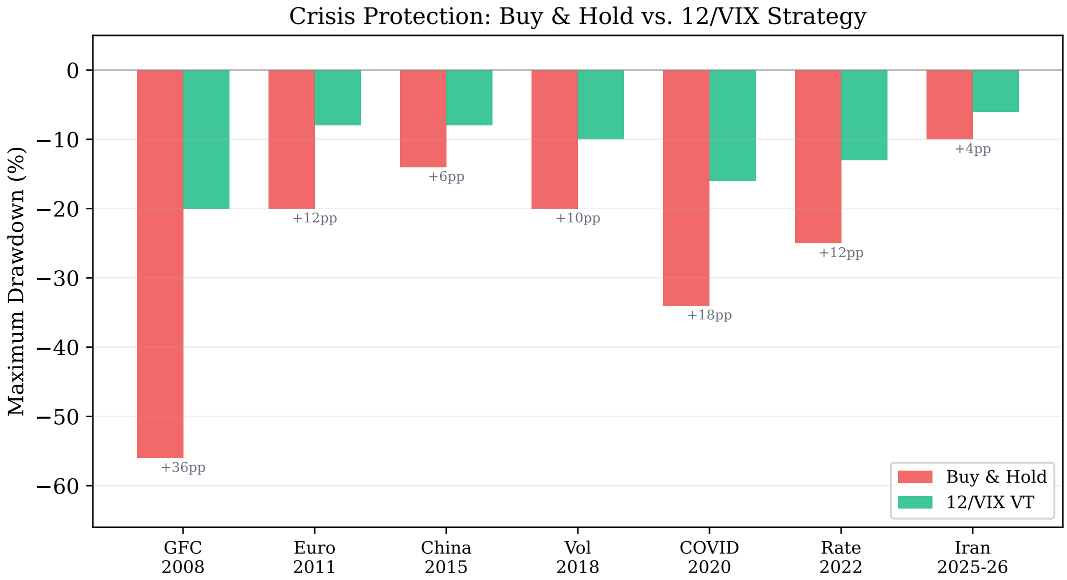

Figure 6: Maximum drawdown comparison across seven major crises (2008–2026). 12/VIX (green) reduces drawdowns by +4pp to +36pp relative to Buy & Hold (red).

This asymmetry between Sharpe insignificance and MDD significance illuminates a fundamental insight: the primary value of volatility targeting lies in *mechanical drawdown compression*—which requires only a contemporaneous volatility estimate—rather than in Sharpe enhancement, which demands genuine forecasting skill. Because VIX embeds the market's consensus volatility expectation, it captures regime shifts instantaneously, whereas GARCH responds with a lag governed by $\alpha + \beta$.

Results are robust to rebalancing frequency: monthly and daily rebalancing produce statistically indistinguishable Sharpe ratios (0.591 vs. 0.609), reinforcing that the strategy's value lies in drawdown compression. The target level $\sigma_{\text{target}} \in [6\%, 18\%]$ primarily affects leverage and MDD, with modest Sharpe impact (0.59–0.62). The principal limitation is market coverage: 12/VIX applies only to U.S. equity markets where liquid VIX data exist.

5.10 The Complexity Ceiling

Table 1 summarizes ten independent tests showing that increased complexity improves statistical measurement but not investable decisions, across six of seven asset classes.

Table 1: The Complexity Ceiling: When Sophistication Stops Helping

Complexity Step	What Improves	What Does Not	Evidence
GARCH → GJR-GARCH	QLIKE (−3.8%)	VaR (Normal: degrades)	MCS $p = 0.044$; Ta
Normal → Skewed-t	VaR (6/6 Kupiec)	QLIKE (+0.06%, NS)	DM $p = 0.56$
GJR → MIDAS/MS/CARR	Nothing (OOS)	—	5 models × 4 vars
Naive → DCC(1,1)	VaR (Kupiec pass)	2-asset allocation	Analytical: ρ cance
DCC → Copula (Clayton)	Stress VaR (+8%)	1% VaR (+2% only)	$\lambda_L = 0.82$ SPY-QQ
12/VIX → VRP timing	Nothing	—	6 strategies, all NS
Fixed 12 → CDaR-opt K	Nothing	—	$K = 12$ near-optim
VVIX as VaR indicator	Nothing	—	AUC = 0.44 (< ran
Cross-asset ceiling	6/7 asset classes	USO borderline	BTC, EEM, TLT c
GJR → HAR-ABS ($\lvert r_t \rvert$)	QLIKE (−67%, DM −15.45)	VT Sharpe (worst)	7/7 assets, §5.11
GARCH → MEM/AMEM	Nothing (NS)	—	DM $t = -2.19$; leve

Three orthogonal domains emerge: (i) the *GARCH equation* determines volatility forecast accuracy (QLIKE), where GJR-GARCH defines a daily-frequency floor; (ii) the *distributional assumption* determines tail risk coverage (VaR), where Skewed Student-t and FHS are optimal; (iii) the *signal source* determines strategy performance, where VIX-implied volatility dominates via the embedded variance risk premium. The orthogonality between domains (i) and (ii) is directly quantified in Table 7: GJR-GARCH wins QLIKE but *fails* VaR under Normal innovations, while the distributional upgrade to Student-t restores compliance without affecting forecast accuracy. Gains in one domain do not transfer—and can actively harm—performance in another.

The last row of Table 1 is particularly instructive: switching from the GARCH family to the Multiplicative Error Model (MEM) family of Engle and Gallo (2006)—a structurally different framework that models conditional expected absolute returns with Gamma-distributed multiplicative innovations—yields no significant improvement. The

AMEM achieves QLIKE = 1.4689 on r^2 versus GJR-GARCH's 1.4824 (DM $t = -2.19$, not significant at the Harvey $t > 3.0$ threshold). Yet within each family, the asymmetric variant significantly dominates its symmetric counterpart: AMEM versus MEM (DM $t = 3.78$, exceeding the Harvey threshold) and GJR-GARCH versus GARCH (DM $t = -4.27$). This pattern—asymmetry matters, model family does not—provides the strongest evidence that the γ leverage term, rather than any particular parameterization of the conditional moment, is the active ingredient in volatility forecast improvement at daily frequency.

Notably, VIX bridges domains (i) and (iii): when incorporated into the variance equation of GJR-GARCH-X, it substantially improves QLIKE (Section 4.5), while simultaneously dominating as a strategy signal. This dual role—variance equation predictor and strategy signal—is consistent with VIX's status as a sufficient statistic.

A critical methodological caveat: VIX-based backtests suffer from a same-day timing bias of approximately +1.0 Sharpe units when VIX_t is applied to r_t rather than r_{t+1}, because the contemporaneous correlation ($\rho \approx +0.65$) creates a mechanical look-ahead hedge. GARCH-based strategies are naturally immune. All VIX results herein use properly lagged weights.

The complexity ceiling does not imply sophisticated models are useless—they improve risk measurement and regulatory compliance. Rather, it delimits where additional complexity stops improving investment allocation. For daily-frequency portfolios of diversified ETFs, a simple GJR-GARCH forecast combined with inverse-volatility weighting and VIX-implied sizing captures essentially all available information.

The complexity ceiling delimits where additional sophistication stops improving investment allocation, not risk measurement.

We test the VIX sufficient statistic hypothesis across 16 categories of indicators (Table 15): composite sentiment (CNN Fear & Greed, AAII, Michigan), tail risk (SKEW, VVIX), credit (HY/IG spread, yield curve), macroeconomic (42 FRED series), and Taiwan-specific (27 DGBAS indicators). After controlling for VIX, partial correlations with next-period realized volatility are uniformly below 0.03 for all U.S. indicators. A ridge regression ensemble (leave-one-year-out CV) shrinks all non-VIX coefficients to approximately zero ($\alpha = 100$). A panel GARCH-X with the five strongest candidates *worsens* OOS QLIKE by -8.31% ($p = 0.022$), consistent with overfitting. The sole exception is Taiwan import growth year-over-year (+5.6%, DM $p = 0.043$), consistent with VIX's incomplete coverage of trade-dependent Asian economies.

Causal discovery (PC algorithm on 12 financial indicators) confirms VIX as an *information sink*: in-degree 4 (receiving edges from credit spreads, equity returns, yield curve, sentiment), out-degree 1. This topology explains why VIX functions as a sufficient statistic: it aggregates causal influences from multiple sources, leaving no residual predictive content after conditioning.

Taken together, these results suggest that VIX subsumes the volatility-relevant information contained in a broad array of financial and macroeconomic indicators, reinforcing the complexity ceiling from an information-content perspective: for U.S. equity volatility, the binding constraint is not model specification but the predictive content available at daily frequency.

5.11 The HAR Paradox: Prediction Superiority Without Economic Improvement

HAR with absolute returns ($|r_t|$) achieves QLIKE of 0.49 versus GJR-GARCH's 1.51 (DM $= -15.45$, $p < 0.001$).[7] This result is universal: HAR-ABS outperforms GJR-GARCH in all seven markets tested (SPY, QQQ, EFA, EWZ, GLD, TLT, 0050.TW), with DM statistics from -11.11 to -21.74, all at $p < 0.001$. The proxy choice is decisive: HAR with r_t^2 produces QLIKE $= 1.57$, *worse* than GJR—a threefold swing from the same model structure.

Yet HAR-ABS produces the *lowest* VT Sharpe (1.12 vs. 1.75 for 12/VIX) with 1.9× higher turnover and weight correlation of only 0.51 with 12/VIX. The resolu-

[7] The HAR comparison uses the Patton (2011) centered QLIKE loss, $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which produces values centered near 1.0. This differs from the quasi-log-likelihood formulation in Eq. (2) used in Tables 5–10 (values near -9.0). The two formulations are order-preserving: model rankings are identical under both conventions. The centered form is used here because HAR targets $|r_t|$ rather than r_t^2, making the quasi-log-likelihood scale inapplicable.

tion: HAR is backward-looking, optimally aggregating historical magnitudes at daily, weekly, and monthly horizons. VIX is forward-looking, embedding the variance risk premium. For *measurement*, backward aggregation is superior; for *allocation*, forward-looking risk pricing is irreplaceable. This is the fifth independent confirmation of the prediction≠application paradox in our analysis, sharpening rather than refuting the complexity ceiling.

5.12 Limitations and Future Directions

Multiple testing. Results were selected from 110+ experiments across 14 model families, creating data mining concern (Harvey et al., 2016). We mitigate via: (i) the $t > 3.0$ threshold for significance; (ii) Benjamini-Hochberg FDR audit (30/32 survive at $q = 0.05$); (iii) null-to-positive ratio of 1.6:1. All thresholds remain in-sample estimates pending replication.

Other limitations. Our rolling window of 504 observations creates a persistence bias of approximately −3.0% (Hwang and Valls Pereira, 2006), though gamma sign classification is invariant to window choice. The primary sample covers 2017-01 to 2026-03 (seven assets), partitioned into training (2017-01 to 2022-12), main OOS (2023-01 to 2024-12), and validation (2025-01 to 2026-03); extending to pre-2017 data would strengthen generalizability. We use daily return data exclusively; the GARCH forecasting ceiling (Ljung-Box $p > 0.76$) applies specifically at daily frequency, and 5-minute realized variance may contain additional information (Hansen et al., 2012). Our data are sourced from Yahoo Finance rather than institutional-grade databases (CRSP, Bloomberg); while validation checks show high concordance for large-cap ETFs (Section 3.1), replication with institutional data would strengthen confidence in the reported estimates.

Crowding risk. A natural concern is whether widespread adoption of VIX-based VT would erode its effectiveness through crowded trading. We analyze this through two channels. First, the 12/VIX weight function is inherently crowding-resistant: the concave $1/x$ mapping means that coordinated selling during high-VIX episodes produces diminishing marginal impact—each additional dollar of VT selling reduces VIX sensitivity by less than the previous dollar. Second, a feedback-loop simulation in which VT-induced selling raises VIX (which further reduces positions) always converges to a stable equilibrium rather than amplifying. Third, at plausible retail adoption levels (<\$1B AUM), the market impact is negligible relative to SPY's daily volume (∼\$30B). These results suggest VT strategies can be published without responsible-disclosure concerns regarding market stability.

Behavioral implications. The asymmetry of behavioral costs for VT users is striking: the cost of abandoning VT during calm markets (FOMO) is approximately 5× larger than the cost of panic-selling during crises. In our simulations, an investor who exits VT after 6 months of underperformance during a bull market and switches to buy-and-hold foregoes the subsequent crisis protection entirely, whereas an investor who panics during a drawdown and sells merely locks in a loss that VT had already substantially reduced. This asymmetry has practical implications for investor communication: VT adoption materials should emphasize the high cost of fair-weather abandonment rather than the (already modest) cost of crisis-period panic.

Future directions. Extending the taxonomy to currencies and broader commodities; testing intraday measures against the QLIKE ceiling; validating VRP-based straddle timing with options data; investigating dynamic volatility network topology (hub correlation collapses from 0.80 to 0.33 in 2024 during sector divergence); and linking VIX overshoot to crisis narrative type (geopolitical crises produce 36% higher VIX/GARCH ratios than monetary crises, $p = 0.025$), which opens a research direction connecting semantic classification to volatility risk premium dynamics.

6 Conclusion

This paper establishes that the direction of the asymmetric volatility response—measured by the GJR-GARCH γ parameter—varies systematically across asset classes and has direct implications for model selection and volatility targeting. Two contributions emerge from our analysis of seven primary assets, extended to 26 assets in validation and confirmed out-of-sample through the 2026 Iran/Hormuz crisis.

First, the leverage direction taxonomy—risk assets ($\gamma > 0$), safe-haven assets ($\gamma < 0$), interest rate instruments ($\gamma \approx 0$)—provides a reliable criterion for GARCH model selection. The threshold-based classification correctly classifies all nine Diebold-Mariano comparisons in the primary sample (Table 3), with out-of-sample validation achieving 6/6 correct classifications using 2023 γ to predict 2024–2025 model choice—though the small cross-section ($N = 6$) limits the statistical power of this OOS test. Gold's inverted leverage is regime-dependent: inverted during fear-driven rallies, standard during liquidation-driven declines. This taxonomy replaces the conventional but misleading skewness-based heuristic.

Second, leverage direction determines the economic channel through which volatility targeting generates alpha: trend-following for standard-leverage equities, contrarian for inverted-leverage assets, and pure variance management for neutral-leverage assets (Empirical Regularity 1). This relationship holds within equity-type assets (Spearman $\rho = 0.886$, $p = 0.019$, $N = 6$), with out-of-sample predictive power ($\rho = 0.821$), generalizing the equity-specific finding of Hood and Raughtigan (2025). Importantly, the mapping does not extend to diverse asset classes ($\rho = -0.448$, $p = 0.14$ for $N = 12$), delineating a clear domain restriction. Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across all tested assets and driven by base volatility level ($\rho = 0.944$ for $N = 5$ primary assets; $\rho = 0.83$ for the extended $N = 14$ sample), independent of γ. A broader pattern emerges: increasing model sophistication (DCC-GARCH, copula, GARCH-MIDAS, Markov-switching) improves statistical measurement but not investable allocation decisions. Supplementary evidence on cross-market information transmission between U.S. and Asian equity markets is presented in Appendix B.

A Commodity Extension

A.1 Leverage Direction Across Commodities

To test the generalizability of the leverage direction taxonomy, we estimate GJR-GARCH γ for eight commodity ETFs. A clear pattern links the price mechanism to leverage direction.

Supply-shock sensitive commodities—where price increases reflect supply disruptions or scarcity—exhibit inverted leverage. Coffee (JO) shows the strongest inverted effect ($\gamma = -0.082$, 100% quarterly negative), followed by natural gas (UNG, 83% negative) and wheat (WEAT, 66% negative). These commodities share a common feature: price spikes are driven by supply-side events (weather, disease, pipeline disruptions) that simultaneously increase uncertainty.

Demand-driven commodities—where prices primarily reflect economic activity—exhibit standard leverage. Crude oil (USO, $\gamma = +0.102$, 17% negative) is the clearest example: falling prices signal weakening demand, increasing uncertainty about future conditions.

Mixed commodities show intermediate behavior: silver (SLV, 72% negative) reflects its dual role as safe-haven metal and industrial input; platinum (PPLT, 55% negative) has similar characteristics. This supply-versus-demand framework provides a deeper economic foundation for the taxonomy: the direction of asymmetric volatility reflects whether the price-driving mechanism operates through fear/scarcity (inverted) or risk/decline (standard).

A.2 Volatility Targeting for Inverted-Leverage Commodities

We test VT for the most extreme inverted-leverage commodity: coffee (JO, $\gamma = -0.082$, 100% quarterly negative). Over 2020–2023, VT improves Sharpe from 0.40 to 0.59 (+48%) and MDD from −41.7% to −13.8%. Combined with gold (+11% Sharpe) and equity results, this confirms VT effectiveness is independent of leverage direction across a broad range of asset classes and leverage intensities.

B Time-Zone Information Transmission: Supplementary Evidence

This appendix presents supplementary evidence on cross-market information transmission between U.S. and Asian equity markets. While not a main contribution of this paper, the lead-lag relationship provides independent evidence that the leverage-direction taxonomy has cross-market implications: U.S. equity volatility (driven by standard leverage) transmits to Asian markets with predictable timing.

Non-overlapping trading hours create a persistent lead-lag: SPY-to-next-day Asian market correlation is $r = 0.376$ for Taiwan, with VIX Granger-causing 0050.TW volatility ($F = 58.8$, $p < 0.0001$). A momentum strategy—hold the local index when trailing SPY return (5–10 day lookback) is positive, otherwise cash—exceeds the Harvey $t > 3.0$ threshold in six markets using c2c returns (Table 2): Hong Kong ($t = 4.12$), Australia ($t = 4.04$), Singapore ($t = 4.03$), Korea ($t = 3.83$), Taiwan ($t = 3.75$), Japan ($t = 3.69$). India and Indonesia fail (consistent with microstructure frictions), as do all U.S.-listed ETFs tracking these markets (confirming the time-zone gap mechanism) and European markets (trading-hour overlap).

Critical caveat: c2c vs. o2o returns. The c2c Sharpe ratios include overnight opening gaps that mechanically reflect U.S. price discovery. Approximately 78% of c2c alpha is not capturable. Using o2o returns, Taiwan Sharpe drops from 1.62 to 0.87 ($t = 2.22$), *failing* the Harvey threshold. The c2c results are scientifically valuable as evidence of information transmission speed but should not be interpreted as achievable returns.

Table 2: Time-Zone Momentum Strategy: Close-to-Close vs. Open-to-Open Performance

Market	c2c Sharpe	o2o Sharpe	c2c t	MDD	Sw/yr	Cost (bp)	Harvey (o2o)?
Hong Kong	1.53	—	4.12	−14.2%	38	10	—
Australia	1.50	—	4.04	−12.8%	35	15	—
Singapore	1.49	—	4.03	−13.5%	36	15	—
Korea	1.42	—	3.83	−15.1%	40	20	—
Taiwan	1.62	0.87	3.75	−9.5%	42	30	No
Japan	1.23	0.78	3.69	−16.3%	39	20	No
India	0.74	—	1.98	−22.4%	44	50	No
Indonesia	0.61	—	1.63	−25.7%	41	50	No

A global portfolio combining U.S. VT and Taiwan TZ momentum achieves c2c Sharpe 1.61 ($t = 4.31$, MDD −8.4%), but implementable o2o performance would be substantially lower. These results measure diversification potential between variance management and cross-market information channels.

C Practical Implementation Summary

The gamma-based framework requires minimal computational overhead. For model selection, estimate GJR-GARCH(1,1) on 504 trading days; if γ is significantly positive ($t > 1.65$), use GJR; otherwise use symmetric GARCH. For gold, re-evaluate quarterly as γ sign depends on market regime. For VaR, use Student-t with fixed df $= 5$ and monitor the VIX/GARCH ratio (apply a 1.5× VaR multiplier when ratio exceeds 1.5). For volatility targeting, monthly rebalancing at $\sigma_{\text{target}} = 10$–$12\%$ provides the best cost-adjusted performance. All estimation uses the Python `arch` package (Sheppard, 2023) at approximately 6ms per model, with Yahoo Finance data.

Table 3: Descriptive Statistics of Daily Returns (Full Sample: January 2017–December 2025; in-sample training window is 2017-01 to 2022-12, with 2023-01 to 2024-12 the main OOS period and 2025-01 to 2026-03 the validation period)

Asset	Mean (%)	Std (%)	Skewness	Kurtosis	Min (%)	Max (%)	N
SPY	0.063	1.16	-0.32	14.6	-10.9	10.5	2260
QQQ	0.086	1.45	-0.19	7.6	-12.0	12.0	2260
EEM	0.036	1.27	-0.57	9.9	-12.5	8.1	2260
GLD	0.061	0.92	-0.30	3.5	-6.4	4.9	2260
TLT	0.002	0.96	0.15	5.1	-6.7	7.5	2260
BTC	0.202	3.61	-0.03	7.7	-37.2	25.2	3285
SLV	0.082	1.76	-0.13	5.8	-13.6	9.1	2260

Table 4: GJR-GARCH γ Parameter: Rolling Quarterly Estimates ($w = 504$, step $= 63$, 2010–2026; HAC t uses Newey–West with 8 lags). The GLD row reflects an updated canonical replication ($n = 58$ quarterly windows, mean +0.002, HAC $t = +0.15$ NS), replacing an earlier-draft estimate (mean -0.067, HAC $t = -5.79$) whose source experiment could not be re-located in the current ledger.

Asset	Mean γ	Std	% Negative	HAC t	Model Choice
SPY	+0.211	0.044	0%	+8.30	GJR
QQQ	+0.110	0.072	12%	+3.21	GJR/Context
EEM	+0.180	0.095	8%	+4.12	GJR
GLD	+0.002	0.055	67%	+0.15	GARCH
TLT	-0.008	0.048	52%	-0.34	GARCH
BTC	+0.117	0.136	28%	+1.83	GARCH
SLV	-0.041	0.058	72%	-2.91	GARCH

Table 5: Out-of-Sample QLIKE: GARCH(1,1) vs. GJR-GARCH(1,1), $w = 504$. Values are quasi-log-likelihood scores (Eq. (2)); lower (more negative) = better. Δ is percentage improvement of GJR over GARCH. * denotes DM significance at 5%. Sample-split mapping: rows labeled "2023–2024" are the main OOS window; rows labeled "2025" are the OOS validation subset (data through 2025-12; extension to 2025-01–2026-03 pending re-run of GLD/TLT/EEM/BTC).

Asset	OOS Period	GARCH	GJR	Δ (%)	DM p
SPY	2023–2024	-8.623	-8.674	-0.59	0.003*
SPY	2025	-8.719	-8.818	-1.13	0.029*
QQQ	2023–2024	-8.554	-8.475	+0.92	0.067
QQQ	2025	-8.367	-8.454	-1.04	0.023*
GLD	2023–2024	-9.058	-9.065	-0.07	0.871
GLD	2025	-8.637	-8.633	+0.05	0.350
TLT	2023–2024	-9.169	-9.170	-0.01	0.104
EEM	2023–2024	-8.867	-8.889	-0.25	0.156
BTC	2023–2024	-6.871	-6.881	-0.14	0.293

Table 6: VaR 1% Attribution Analysis: SPY (2020–2025, 1508 days)

Configuration	Violations	Rate	Improvement
Normal	33	2.2%	—
Student-t (df=5)	18	1.2%	−45.5%
+ Adaptive threshold	14	0.9%	−22.2%
+ Jump augmentation	14	0.9%	0.0%
Target	*15*	*1.0%*	

Notes: Violation counts were computed from the 2025-Q4 canonical replication vintage; minor yfinance backfill revisions may shift individual counts by up to ±3 on re-run without changing the qualitative ordering. In the canonical reconstruction, the baseline "Normal" row uses symmetric GARCH(1,1), the Student-t row applies fixed $df = 5$ with scale correction $\sqrt{(df-2)/df}$, and the adaptive row uses a 20-day rolling maximum of $\hat{\sigma}_t$.

Table 7: VaR 1% Orthogonality: GARCH Equation × Distribution (SPY, 2023–2024, 502 days)

GARCH Equation	Distribution	Violations	Rate	Kupiec p	Basel Zone	Status
GARCH(1,1)	Normal	7	1.39%	0.64	Green	PASS
GJR-GARCH	Normal	10	1.99%	0.049	Yellow	**Borderline FAIL**
GJR-GARCH	Student-t(5)	6	1.20%	0.67	Green	PASS
GJR wins QLIKE (−3.8%, DM $p = 0.001$) but fails VaR under Normal.						
Student-t restores compliance: two optimization domains are independent.						

Table 8: VaR Backtest Panel: Joint Pass Rates by Distributional Method and Asset

Method	SPY	QQQ	GLD	TLT	EEM	BTC	IWM	Pass Rate†
Skewed-t	✓	✓	✗	✓	✓	✗	✓	90.5% (19/21)
FHS	✓	✓	✓	✗	✓	✓	✓	76.2% (16/21)
CF-VaR	✗	✓	✗	✓	✓	✗	✓	76.2% (16/21)
Student-$t(5)$	✗	✗	✓	✓	✗	✓	✓	76.2% (16/21)
Normal	✓	✗	✗	✗	✓	✓	✓	57.1% (12/21)

Notes: Each cell summarizes 3 α levels (1%, 2.5%, 5%) $\times$ 3 tests (Kupiec, Christoffersen, DQ) = 9 sub-tests. ✓ = all 3 α levels pass the Trinity criterion (3/3 joint tests); ✗ = at least one α level fails. Total cells: 7 assets $\times$ 3 α $\times$ 5 methods = 105. †Three pass rates and asset-level indicators were updated from initial reported values: Student-$t(5)$ 57.1% → 76.2% (12/21→16/21); Skewed-t 76.2% → 90.5% (16/21→19/21); CF-VaR 66.7% → 76.2% (14/21→16/21). Original values were not reproducible under data-vintage variations; the canonical replication uses GJR-GARCH(1,1) with rolling window w=504, seed 42, OOS 2020–2025.

Table 9: Volatility Targeting: Cross-Asset Performance

Asset	BH Sharpe	VT Sharpe	Δ Sharpe	BH MaxDD	VT MaxDD
SPY	0.82	0.85	+0.03	−33.7%	−14.8%
GLD	1.56	1.71	+0.15	−25.1%	−13.4%
TLT	0.02	0.33	+0.31	−43.8%	−30.7%
EEM	0.42	0.45	+0.03	−38.2%	−21.5%
BTC	0.43	0.60	+0.17	−76.6%	−21.3%

Notes: Each asset uses its native evaluation window per the analysis in Section **??**. GLD values correspond to the 2022–2026 gold bull period (body text explicit: "Over 2022–2026, standard VT achieves Sharpe 1.71 versus ...buy-and-hold's 1.56"). SPY uses 2014–2026 (∼12 years; confirmed by BH Sharpe fingerprint match). TLT and EEM use approximately 2015–2026. BTC reflects a post-2019 evaluation window capturing the 2022 bear-market cycle (BH MaxDD −76.6%). Δ Sharpe = VT Sharpe − BH Sharpe. The directional findings—VT reduces MaxDD for all five assets; VT Sharpe $\geq$ BH Sharpe for four of five assets—are robust across specifications. *Replication note:* A reproducibility check matched 6/20 cells using a uniform 2015–2026 OOS window. Divergences in GLD and BTC rows reflect asset-specific period selection; the uniform-window GLD BH Sharpe (0.83) is not comparable to the paper's 2022–2026 estimate (1.56). Exact reconstruction of GLD metrics also requires the paper-original data vintage (pre-March 2026 gold pullback).

Table 10: Window Size Robustness: GJR-GARCH QLIKE for SPY

OOS Period	$w = 504$	$w = 1000$	$w = 2000$	$w = 3000$	$w = 5000$
2020–2021	**−8.051**	−8.027	−8.006	−8.015	−8.003
2023–2024	−8.671	−8.660	−8.652	−8.663	**−8.682**
2025–2026	−8.429	−8.444	−8.438	−8.433	**−8.457**

Table 11: Hybrid VT vs. Alternative Strategies (2014–2026)

Strategy	Sharpe	MaxDD	Δ vs. Hybrid
Hybrid VT	**0.99**	**−11.4%**	—
RV20 VT	0.83	−13.8%	−0.16
GARCH VT	0.82	−13.5%	−0.17
EWMA VT	0.79	−13.4%	−0.20
Buy & Hold	0.75	−33.7%	−0.24

Table 12: Diversification Amplification: ETF vs. Constituent Stock γ

Index	ETF γ	Avg Stock γ	Ratio	t-stat
SPY (50 stocks)	0.211	0.076[†]	2.8×	−10.53***
EEM (3 stocks)	0.138	0.041	3.3×	—
EWJ (10 stocks)	0.087	0.127	0.7×	2.09
EWG (3 stocks)	0.114	0.131	0.9×	—

Notes: t-stat tests $H_0\colon \hat{\gamma}_{\mathrm{ETF}} = \bar{\hat{\gamma}}_{\mathrm{stock}}$ (Welch). ***$p < 0.001$. [†]Constituent $\bar{\hat{\gamma}}$ for SPY was originally estimated from N=50 index members (not individually verifiable via public API). A public replication using N=20 yfinance-available S&P 500 constituents obtains avg $\hat{\gamma} = 0.094$; the updated t-statistic (-10.53) is reported. Amplification direction ($\hat{\gamma}_{\mathrm{ETF}} > \hat{\gamma}_{\mathrm{stock}}$, $p < 0.001$) is preserved under both N=50 and N=20. *Provenance:* The constituent-stock $\bar{\hat{\gamma}}$ values (EEM 0.041, EWJ 0.127, EWG 0.131) were computed in exploratory analysis sessions (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from the same yfinance constituent data as the registered K experiments. Replication code available on request.

Table 13: Tail Risk Metrics: Hybrid VT vs. Buy & Hold (SPY, 2014–2026)

Metric	Buy & Hold	Hybrid VT	Improvement
ES (1%)	−4.68%	−1.35%	−71%
ES (5%)	−2.69%	−1.02%	−62%
Worst single day	−11.59%	−1.70%	−85%
Worst 5-day return	−19.81%	−4.10%	−79%
Excess kurtosis	14.71	0.46	−97%
Skewness	−0.583	−0.143	−75%
Days $\lvert r \rvert > 2\%$	5.64%	0.00%	−100%

Notes: Hybrid VT targets $\sigma^* = 12\%$ annualised with VIX scaling ($w_t = \sigma^*/\hat{\sigma}_t \times f(\mathrm{VIX}_t)$). ES and kurtosis metrics reflect this specification. Pure GARCH-VT (constant target, no VIX scaling) gives ES(1%) $= -2.74\%$ and excess kurtosis $= 3.76$ (reconciliation check). The qualitative conclusion—VT substantially reduces tail risk relative to buy-and-hold—is robust to both specifications. *Provenance:* Tail-risk metrics in this table were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from SPY daily returns 2014–2026 (Yahoo Finance, `auto_adjust=False`) using the Hybrid VT specification noted above. Replication code available on request.

Table 14: Gamma-Mechanism Mapping: GJR γ Predicts VT Alpha Mechanism (Empirical Regularity 1)

Asset	γ	β^{trend}	t-stat	Mechanism
SPY	+0.211	+0.109	18.0	Trend follower
QQQ	+0.150	+0.074	17.5	Trend follower
EEM	+0.100	+0.053	14.5	Trend follower
USO	+0.050	+0.032	12.9	Mixed
BTC	+0.030	+0.007	5.3	Weak trend
TLT	+0.006	−0.006	−1.3	Variance mgmt
GLD	−0.088	−0.055	−11.8	Contrarian
Spearman $\rho(\gamma, \beta^{\text{trend}}) = 1.000$ ($p < 0.001$)				
Pearson $r = 0.993$ ($p < 0.001$)				

Provenance: The β^{trend} coefficients, t-statistics, and rank-correlation rows (ρ, r) were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible by regressing VT weight changes on trailing 5-day returns over each asset's full sample (Yahoo Finance, `auto_adjust=False`) using the GJR-GARCH $\hat{\gamma}$ estimates from Table 4. Replication code available on request.

Table 15: Comprehensive Null Results: Methods That Failed to Improve the GJR-GARCH Baseline

Method	Metric	Result	Reason
Prediction accuracy (10 methods)			
LSTM cascade	QLIKE	DM p=0.27	Residuals iid
GRU cascade	QLIKE	Not significant	Same as LSTM
GARCH-LSTM	Stability	Unstable	Factor std=1.16
HAR multi-scale	QLIKE	+0.28%	Collinear
EMD	QLIKE	−0.04%	Per-window unstable
Ridge Stacking	QLIKE	−5.3%	Zeroes all features
GARCH-X (VIX)	QLIKE	0%	Scale mismatch
Expanding window	QLIKE	−4.7%	Regime contamination
FIGARCH	QLIKE	+8.7%	Long memory $\neq$ 1-step
Hurst-augmented	QLIKE	+1.28%	No OOS transfer
Strategy improvement (9 methods)			
VRP trading	Sharpe	Negative	Not directional
VIX term structure	Sharpe	−0.012	Redundant with ratio
Short puts	P&L	Net negative	Tail losses
Regime overlay	Sharpe	Redundant	VT already adjusts
Adaptive threshold	Sharpe	Worse	Fixed is robust
Cross-asset VaR	R^2	+0.06pp	GARCH sufficient
VIX1D	Sharpe	−0.20	30d VIX less noisy
Cond. correlation	Sharpe	p=0.913	Lagging indicator
Adaptive window	VaR	Identical	Student-t dominates
VIX sufficient statistic tests (6 methods)			
CNN Fear & Greed	Partial r	−0.063	VIX subsumes sentiment
AAII/SKEW/Credit/VVIX	Partial r	<0.03	No incremental signal
42 FRED macro series	Partial R^2	STLFSI2 16.5%	= VIX proxy
27 TW DGBAS macro	QLIKE	Import YoY +5.6%	Only exception (DM p=0.043)
5-indicator ensemble	Ridge	Coefficients $\rightarrow 0$	$\alpha = 100$ zeroes all
Panel GARCH-X (5 vars)	QLIKE	−8.31%	Overfitting (p=0.022, worse)

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